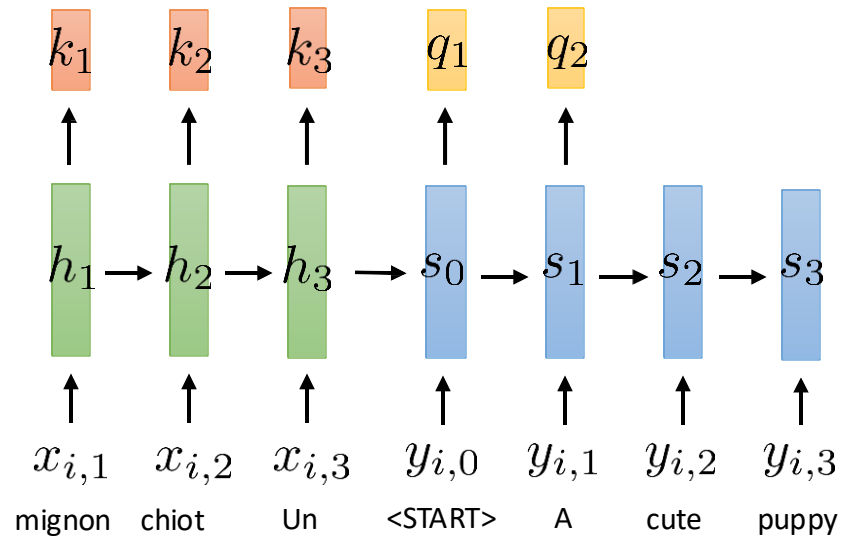


Transformer

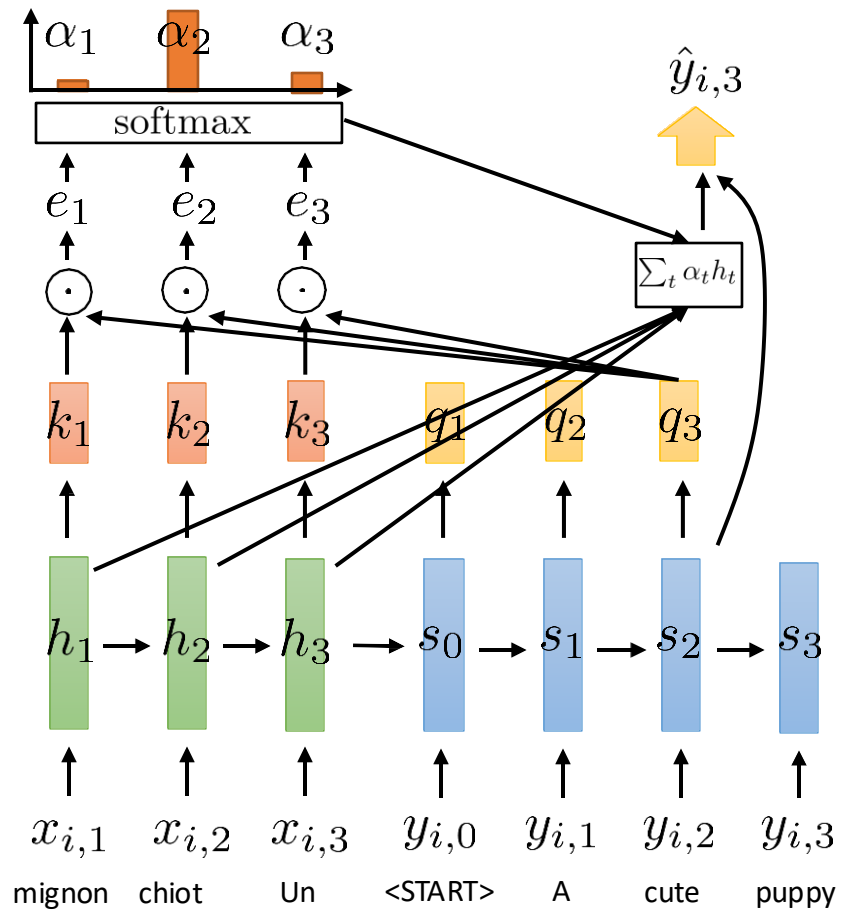
Tanmoy Chakraborty
Associate Professor, IIT Delhi
<https://tanmoychak.com/>

Is Attention All We Need?

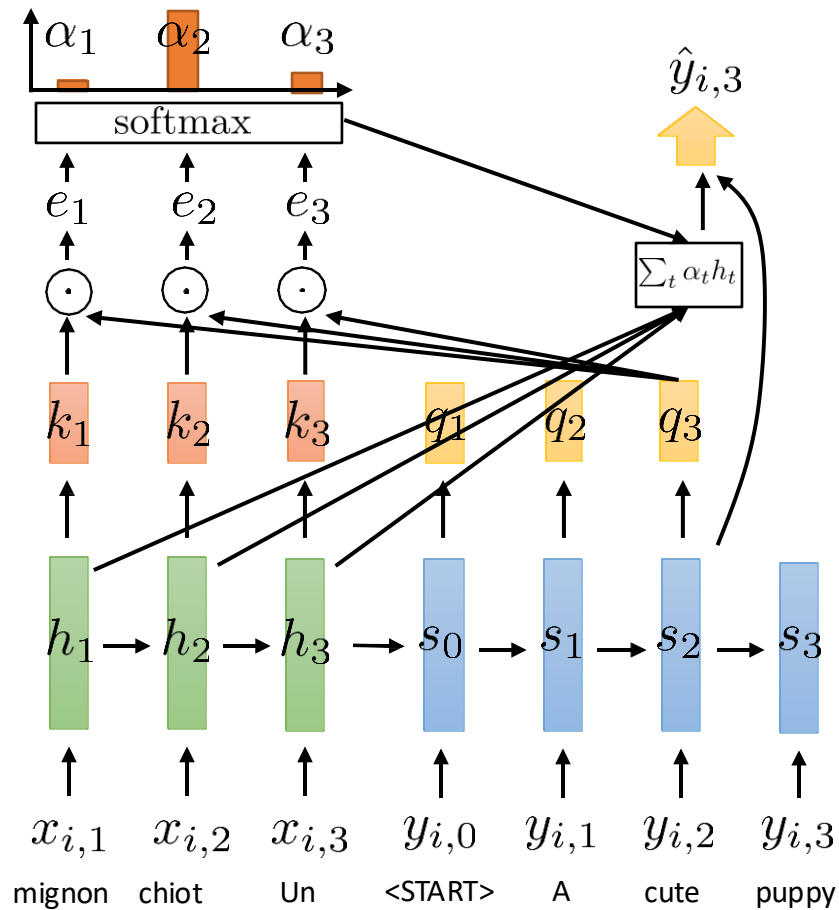
Recap: Attention



Recap: Attention

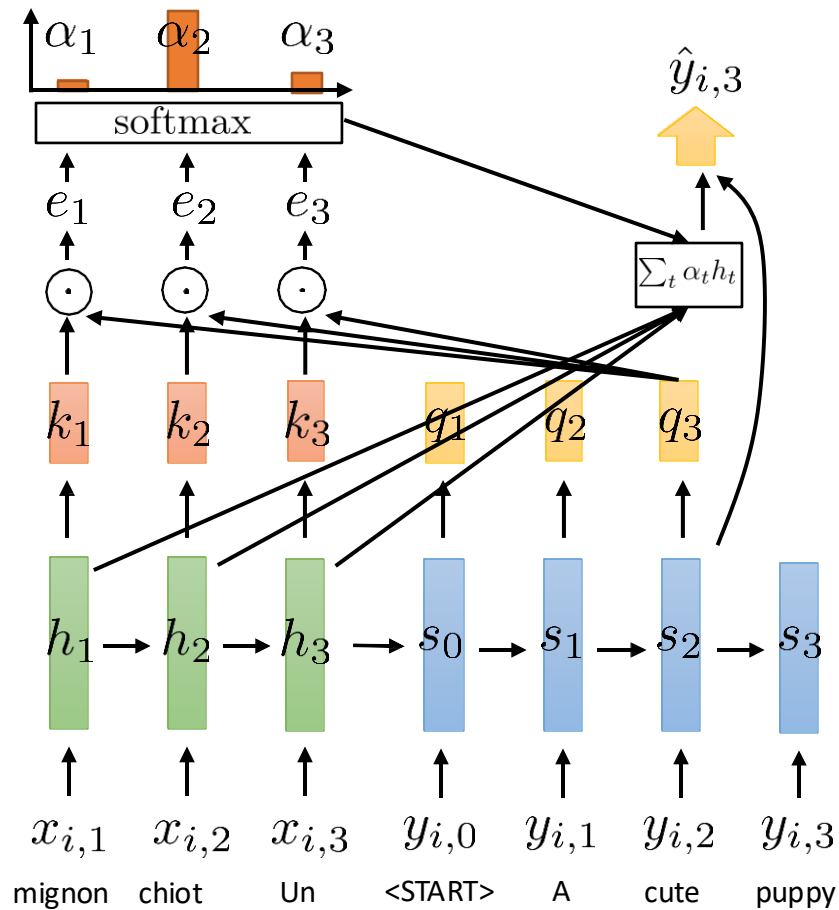


Recap: Attention



- If we have **attention**, do we even need recurrent connections?
- Can we transform our RNN into a **purely attention-based model**?
- Attention can access all time steps simultaneously, potentially doing everything that recurrence can, and even more. However, this approach presents some challenges:

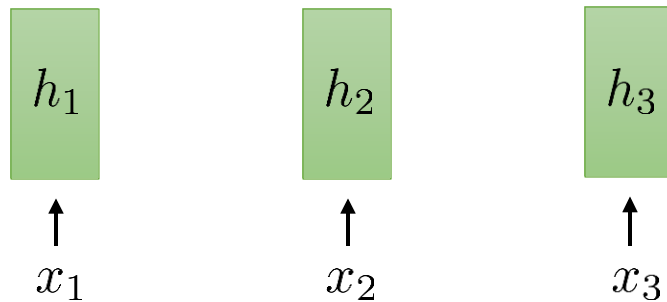
Recap: Attention



- If we have **attention**, do we even need recurrent connections?
- Can we transform our RNN into a **purely attention-based model**?
- Attention can access all time steps simultaneously, potentially doing everything that recurrence can, and even more. However, this approach presents some challenges:

The encoder lacks temporal dependencies at all!

Self-Attention



this is *not* a recurrent model!

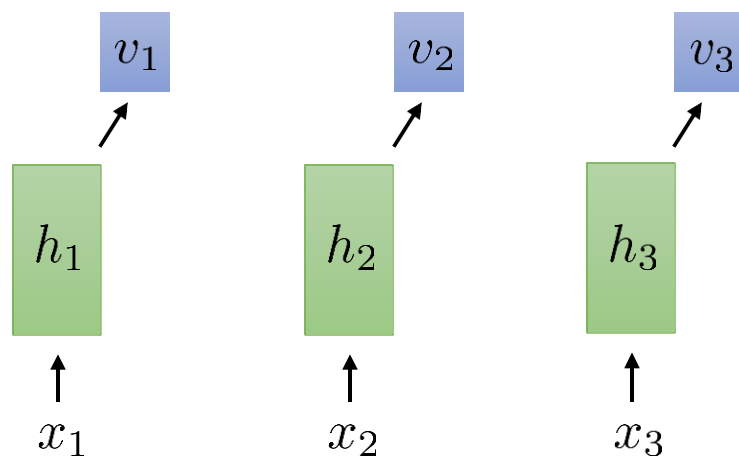
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

← shared weights at all time steps

(or any other nonlinear function)

Self-Attention



this is *not* a recurrent model!

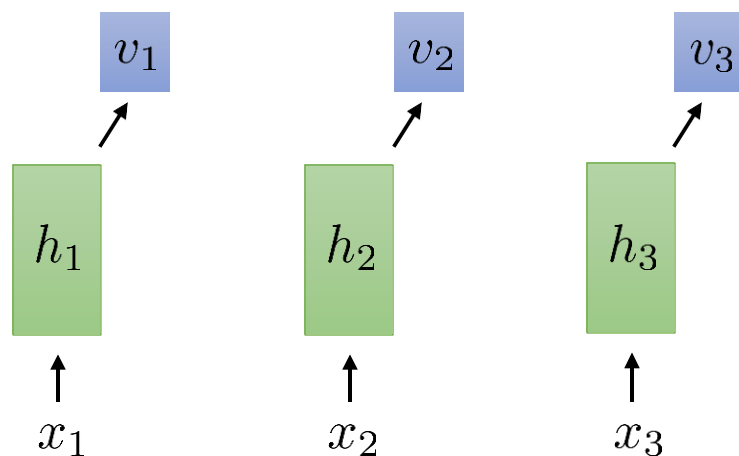
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

this is *not* a recurrent model!

but still weight sharing:

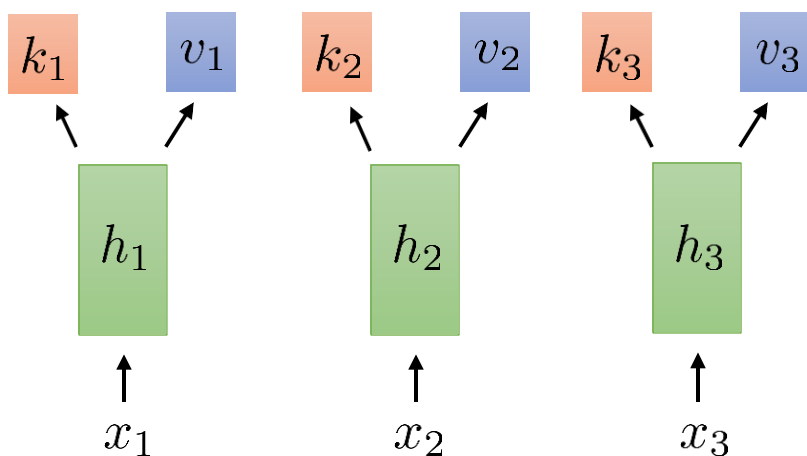
$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$



this is *not* a recurrent model!

but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$
 $k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

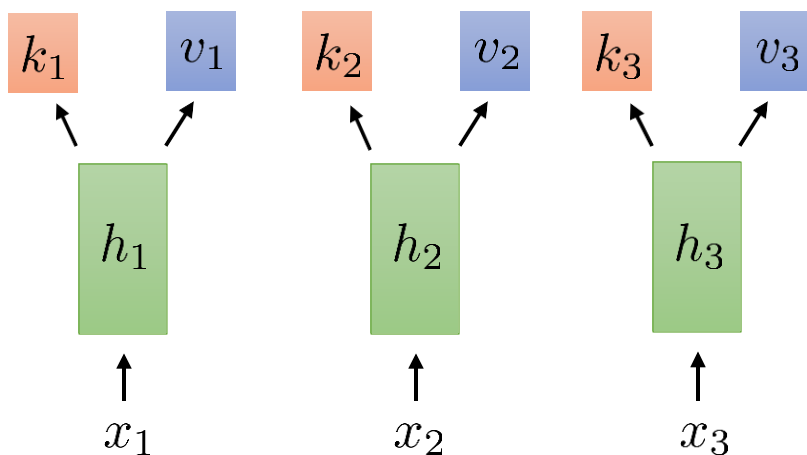
this is *not* a recurrent model!

but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

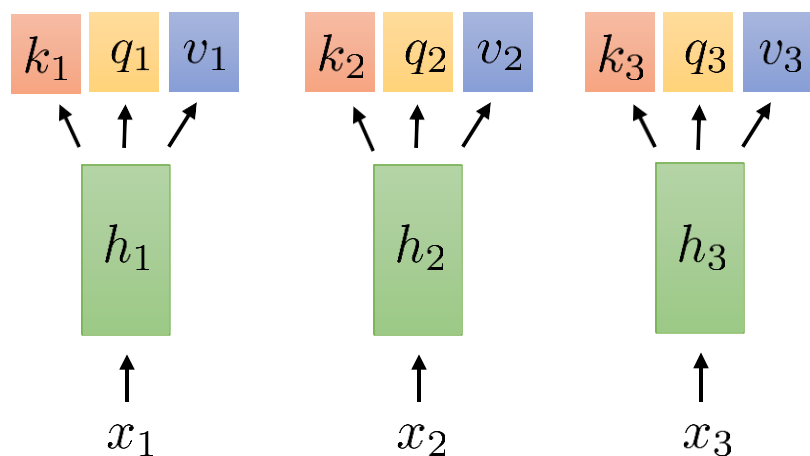
← shared weights at all time steps

(or any other nonlinear function)



Self-Attention

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$
 $k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$



this is *not* a recurrent model!

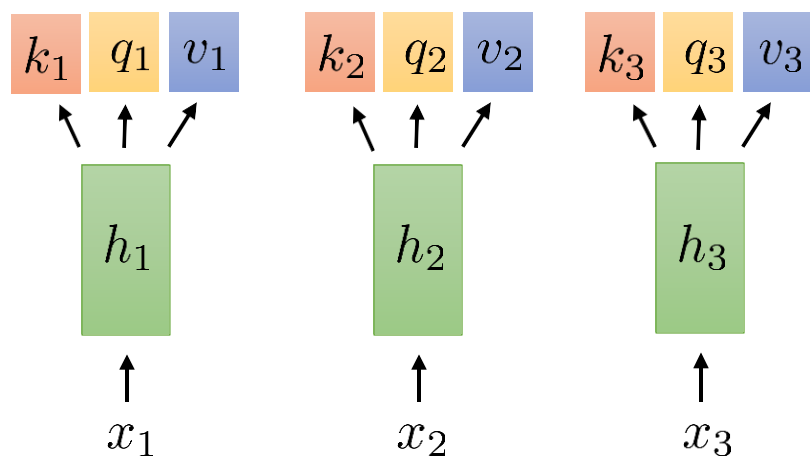
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

← shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

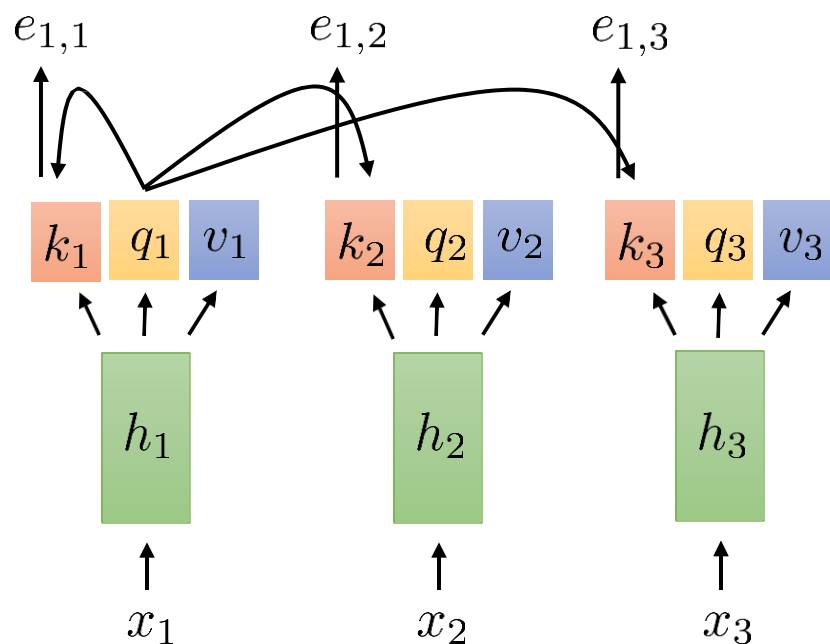
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

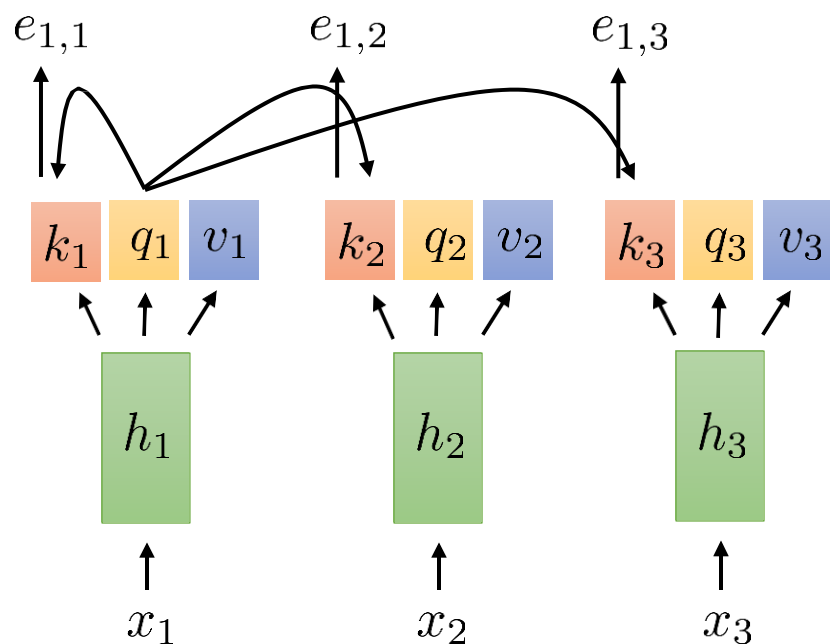
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

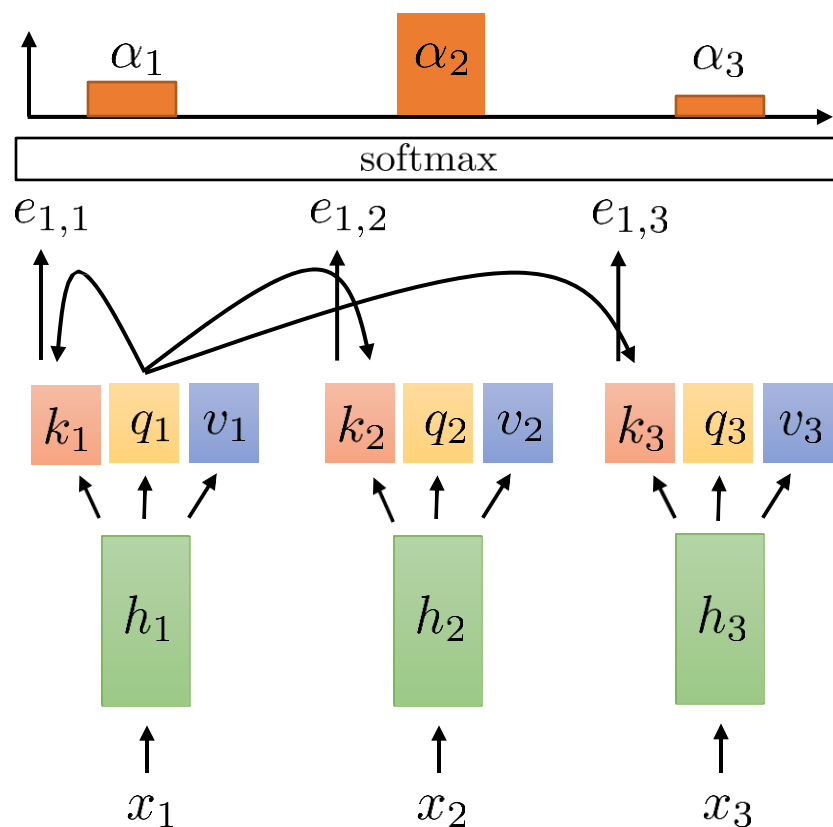
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

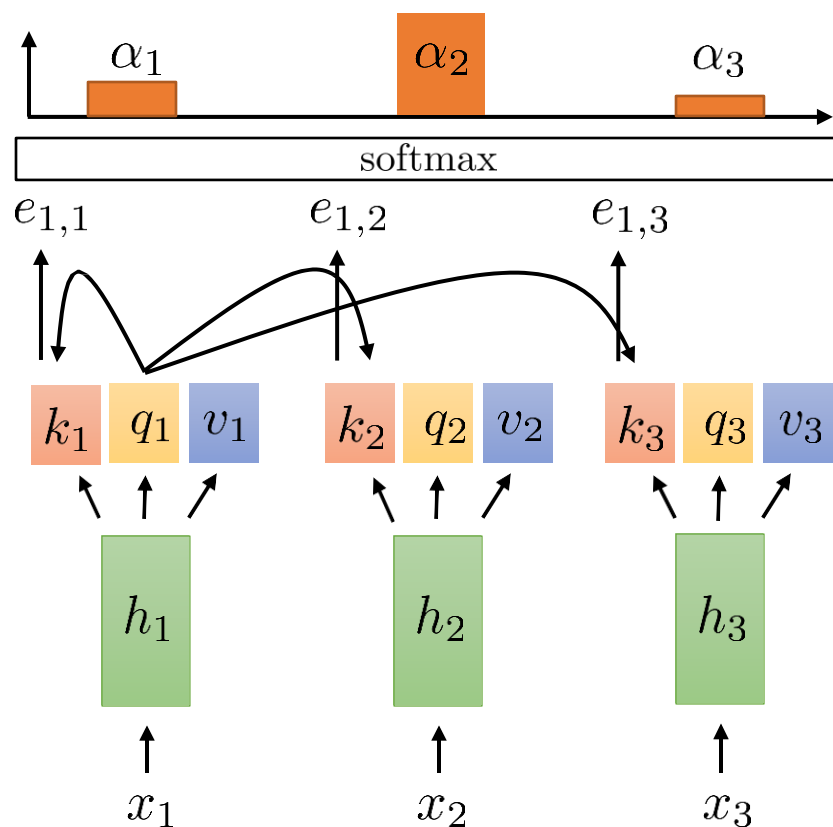
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

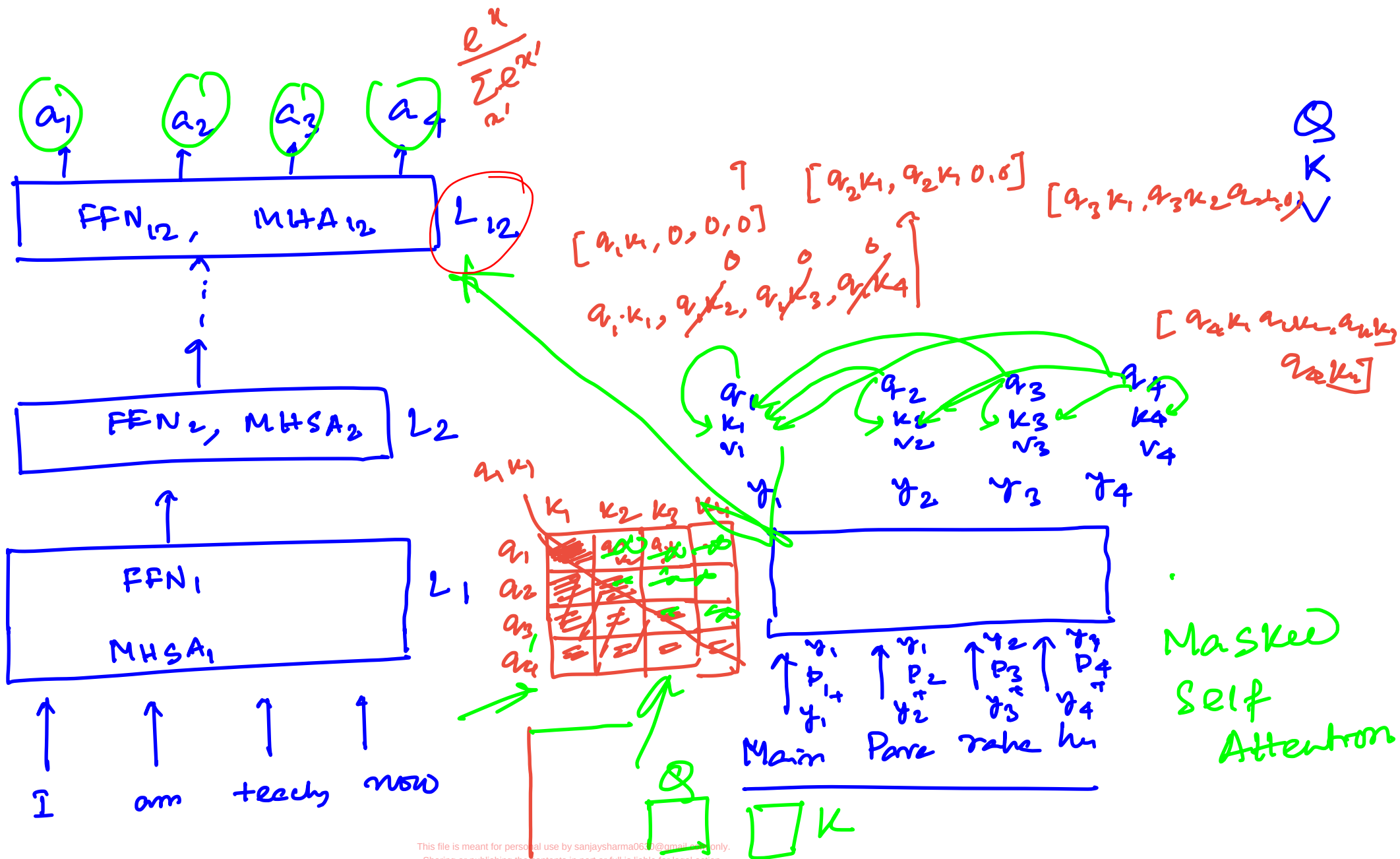
shared weights at all time steps

(or any other nonlinear function)

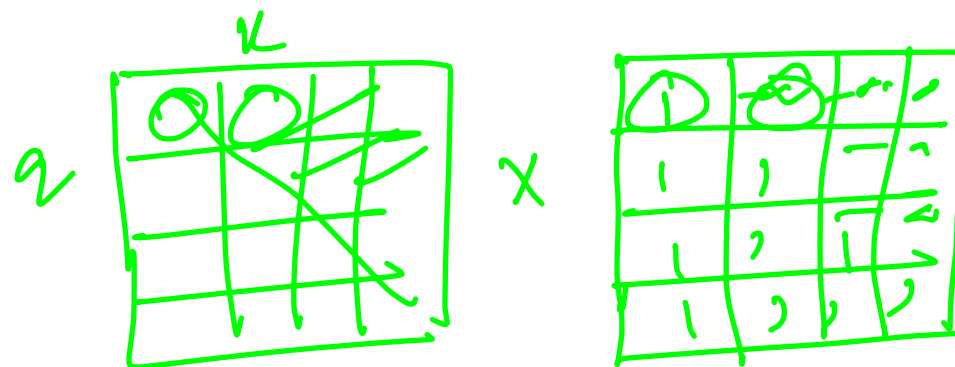
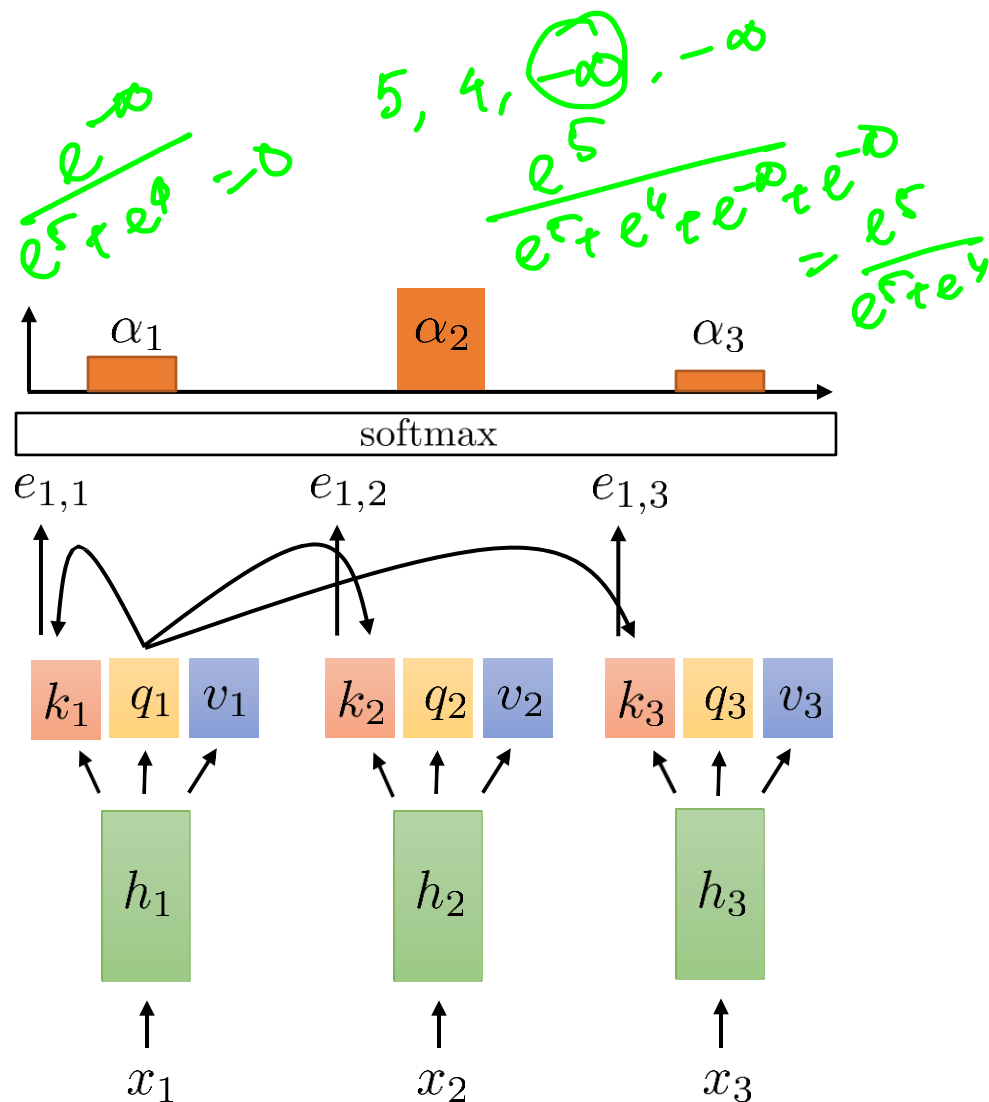
MT

Encoder

EN



Self-Attention



$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

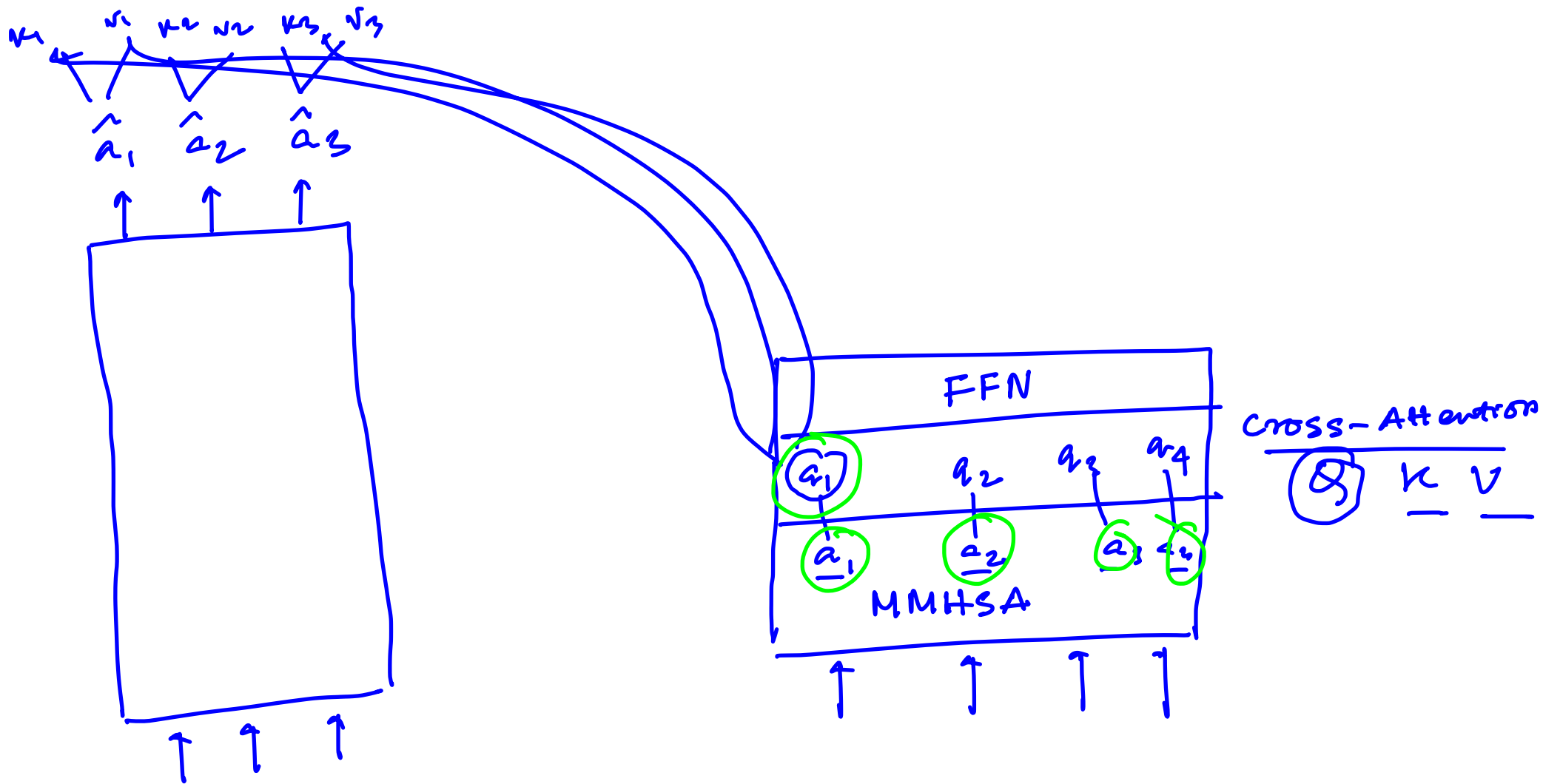
this is *not* a recurrent model!

but still weight sharing:

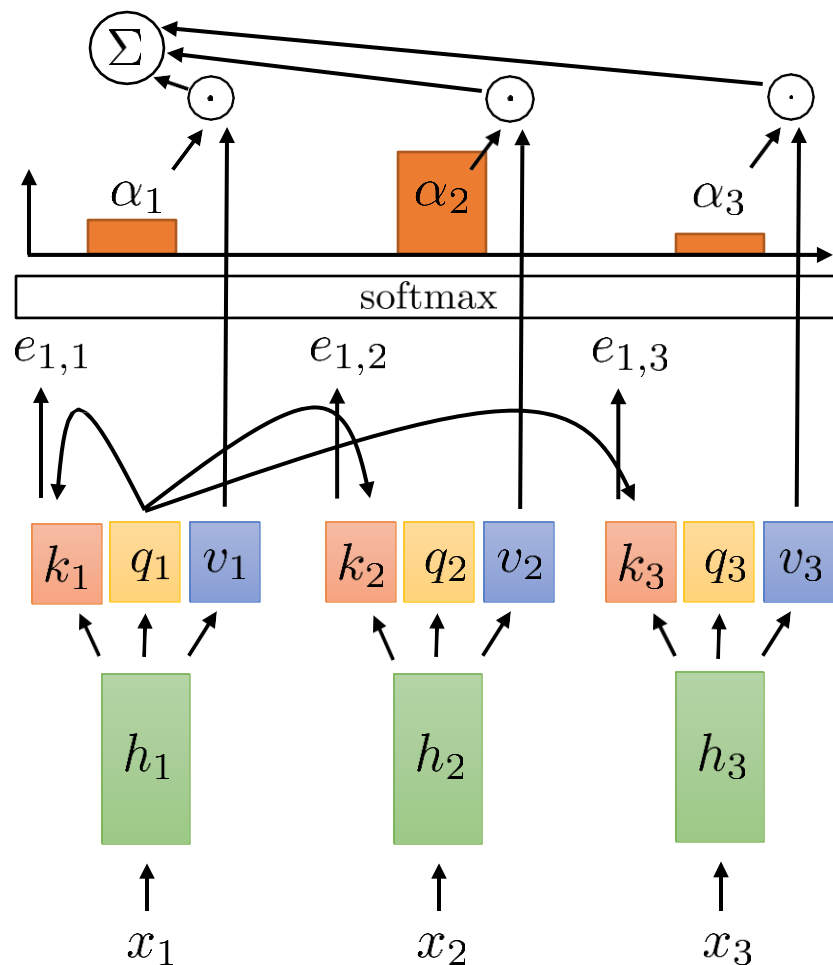
$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)



Self-Attention



$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

ROPE Rotary Positional Encoding

b_1, b_2, b_3, b_4



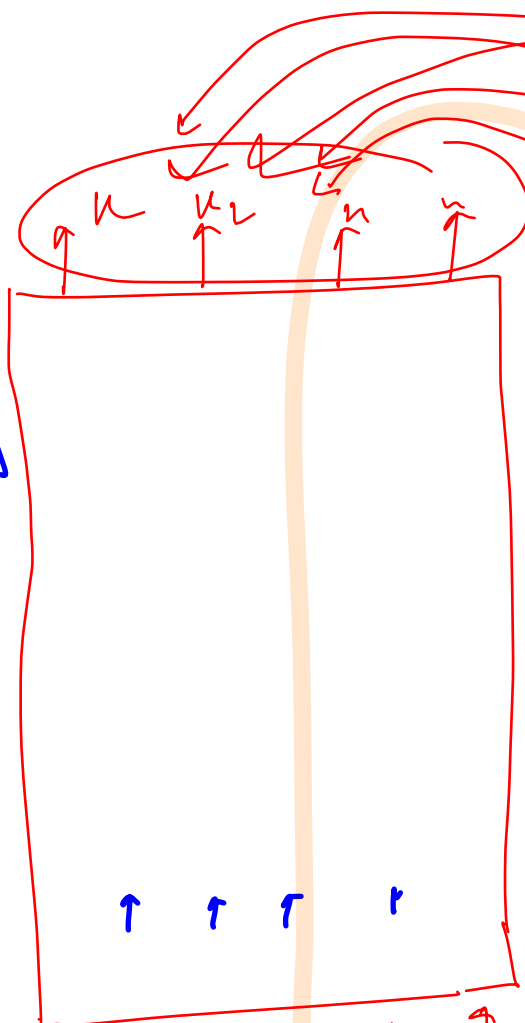
β_j
 x_i

I

am

the

now



CE {



a_1

a_2

a_3

a_4

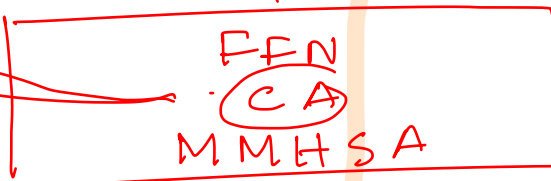
a_5

← Activation
vectors.

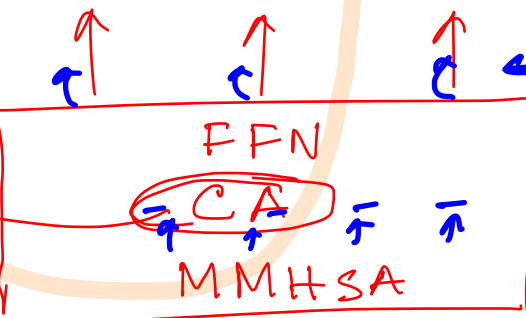
← Activation
vectors.



L_{12}



L_2

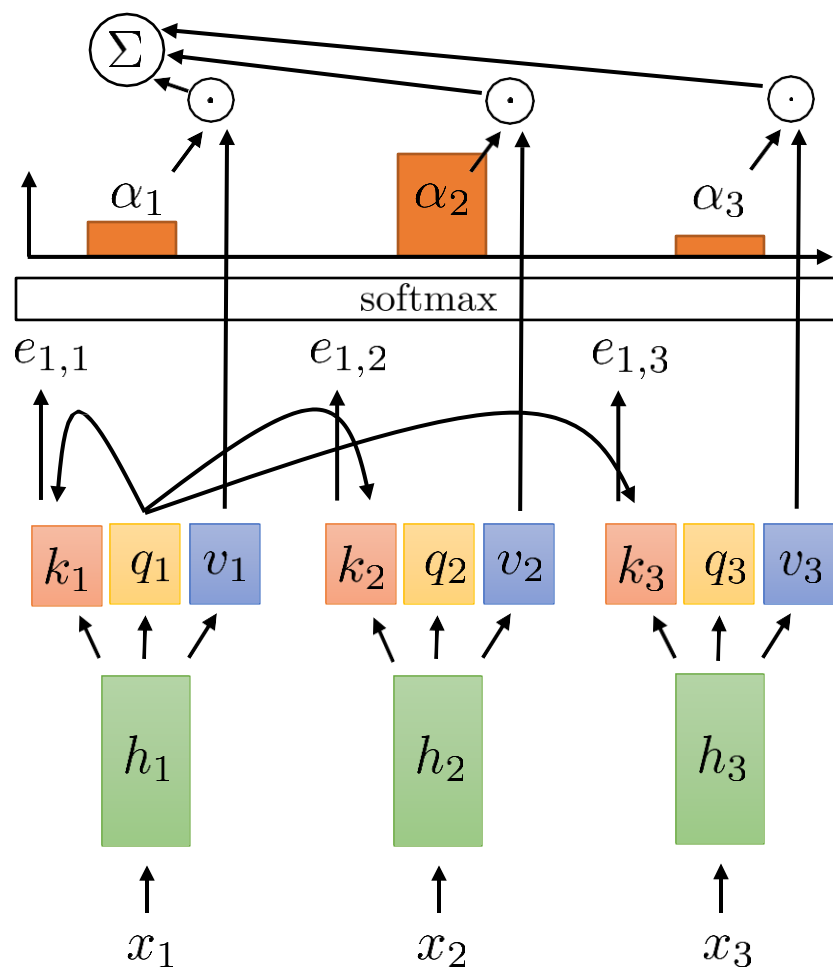


L_1

← activation
vectors

$\langle \text{start} \rangle$ Main Para rch in

Self-Attention



$$a_l = \sum_t \alpha_{l,t} v_t$$

$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

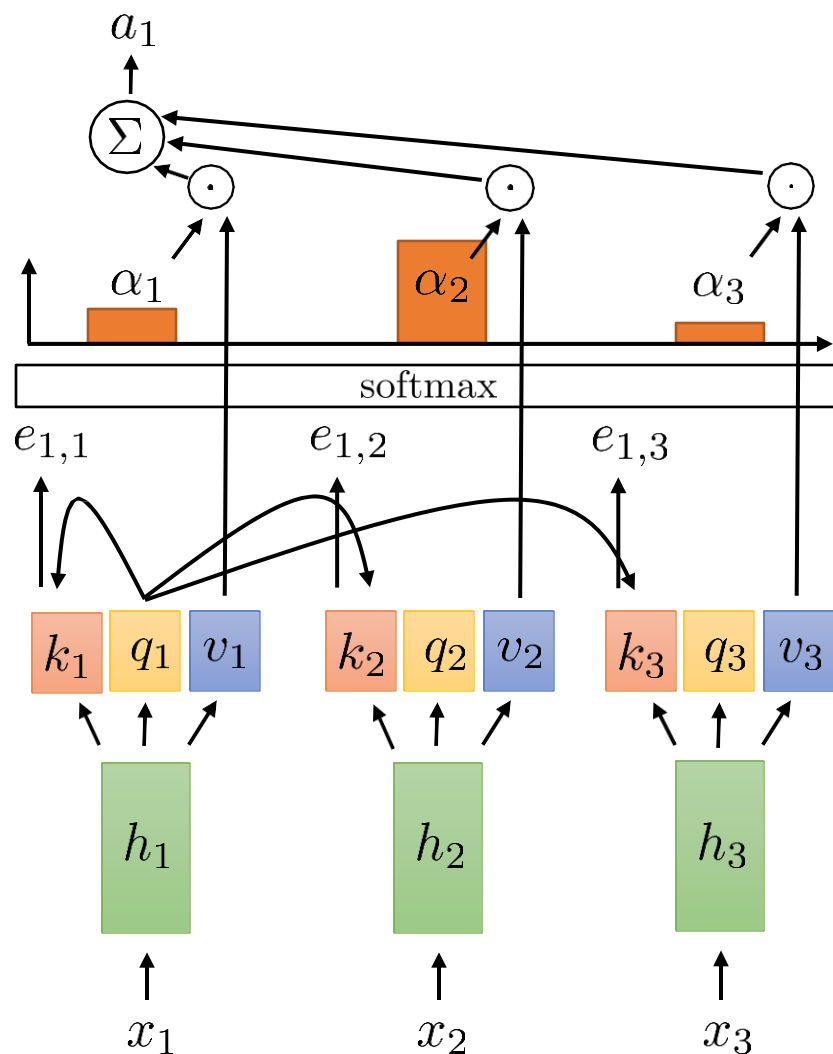
but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention



$$a_l = \sum_t \alpha_{l,t} v_t$$

$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

$v_t = v(h_t)$ before just had $v(h_t) = h_t$, now e.g. $v(h_t) = W_v h_t$

$k_t = k(h_t)$ (just like before) e.g., $k_t = W_k h_t$

$q_t = q(h_t)$ e.g., $q_t = W_q h_t$

this is *not* a recurrent model!

but still weight sharing:

$$h_t = \sigma(Wx_t + b)$$

shared weights at all time steps

(or any other nonlinear function)

Self-Attention

Open AI

2018 → GPT - Decoder only model

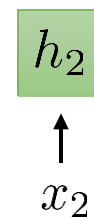
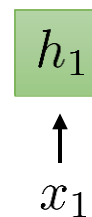
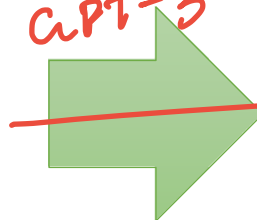
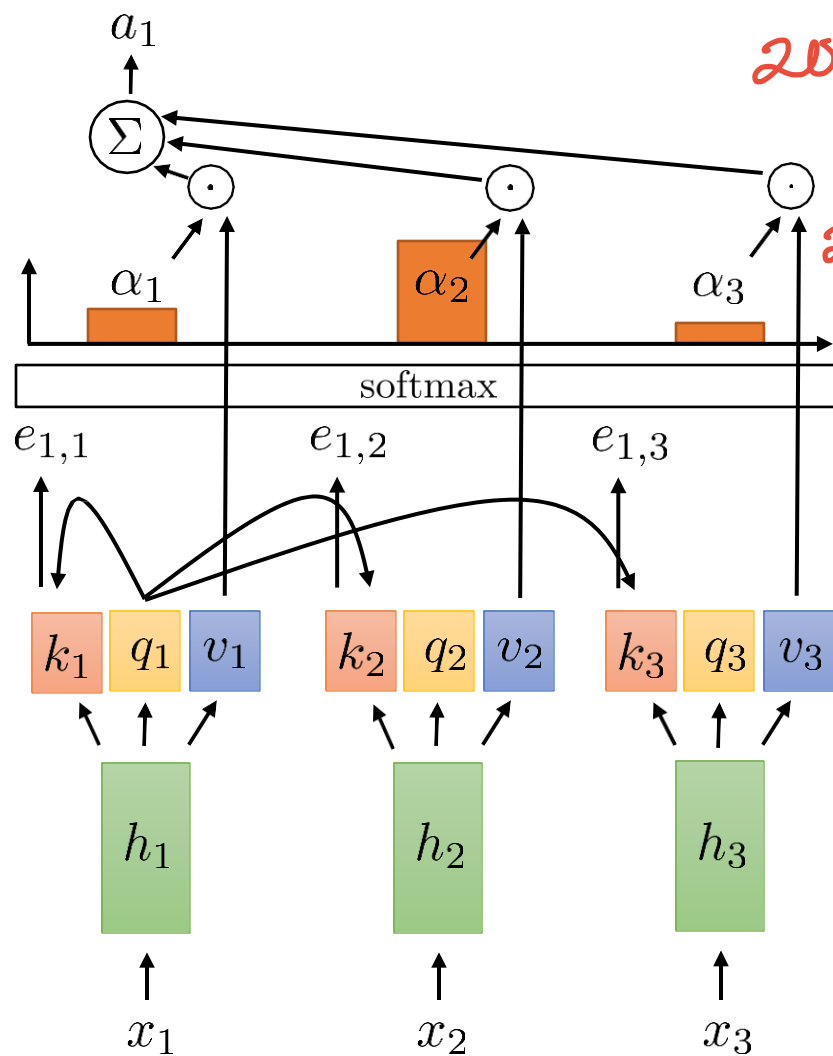
2019 GPT-2

2020 GPT-3

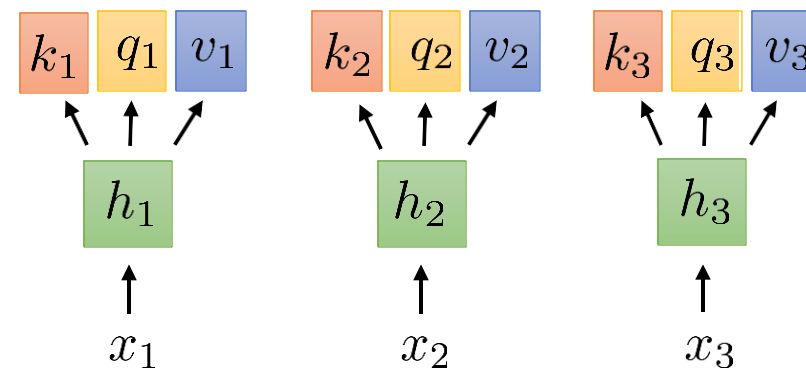
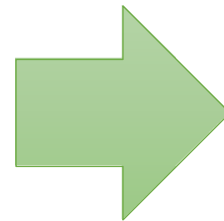
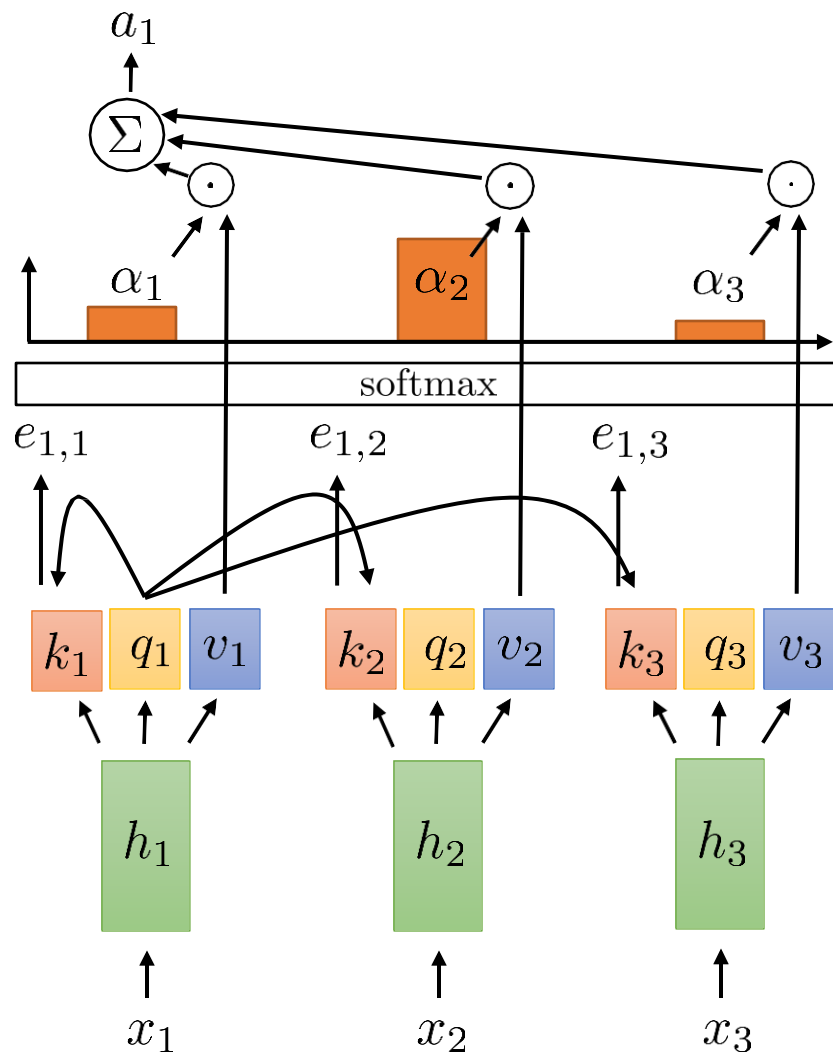
2017 - Transformer

2018 - BERT - Encoder only model

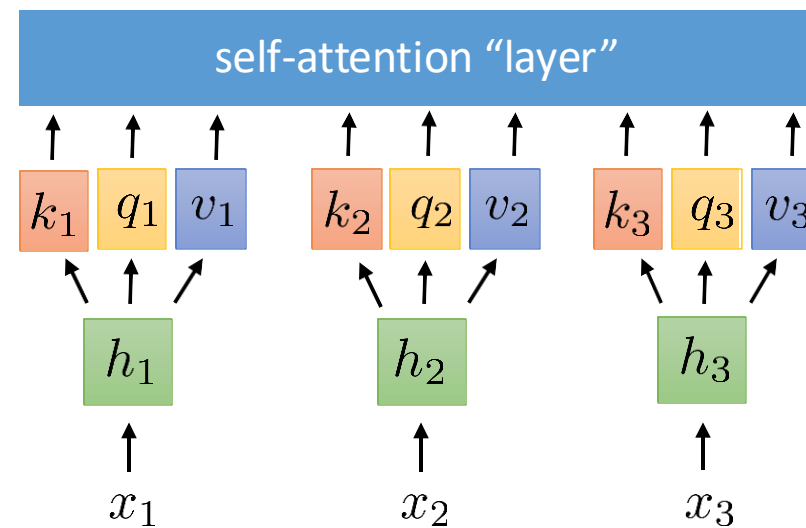
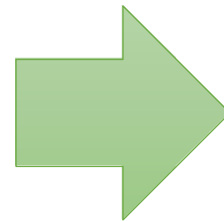
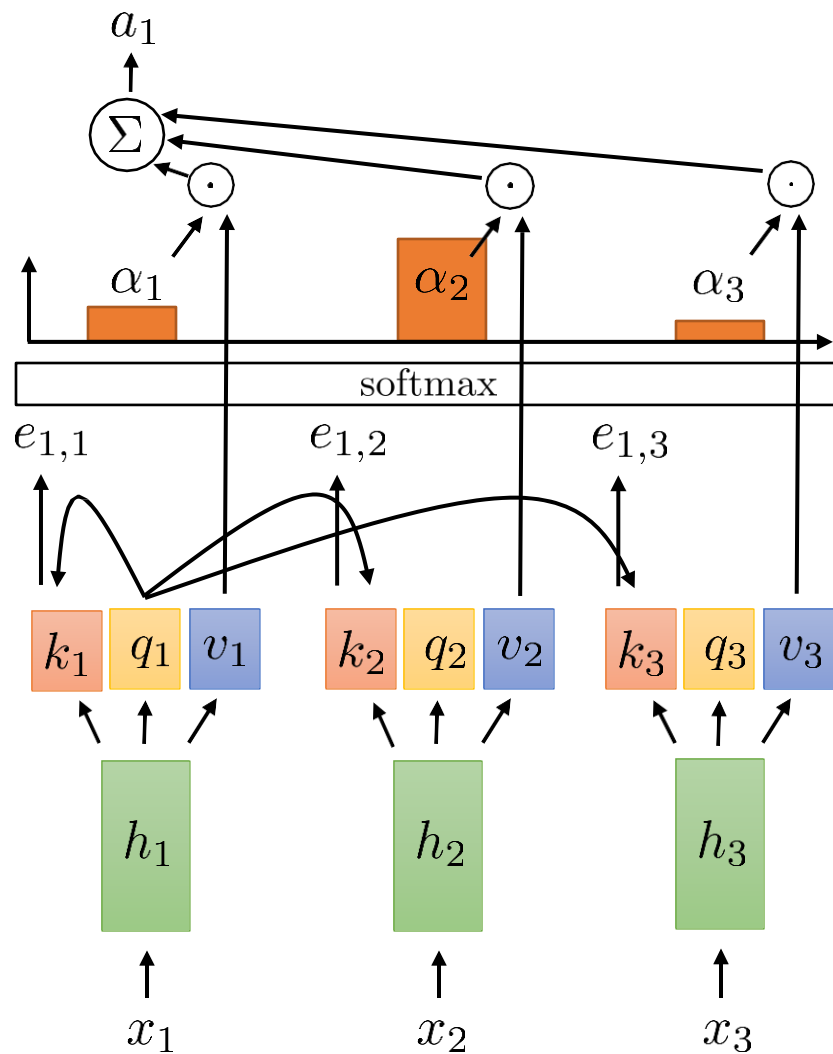
2020 - T5 - En-de model



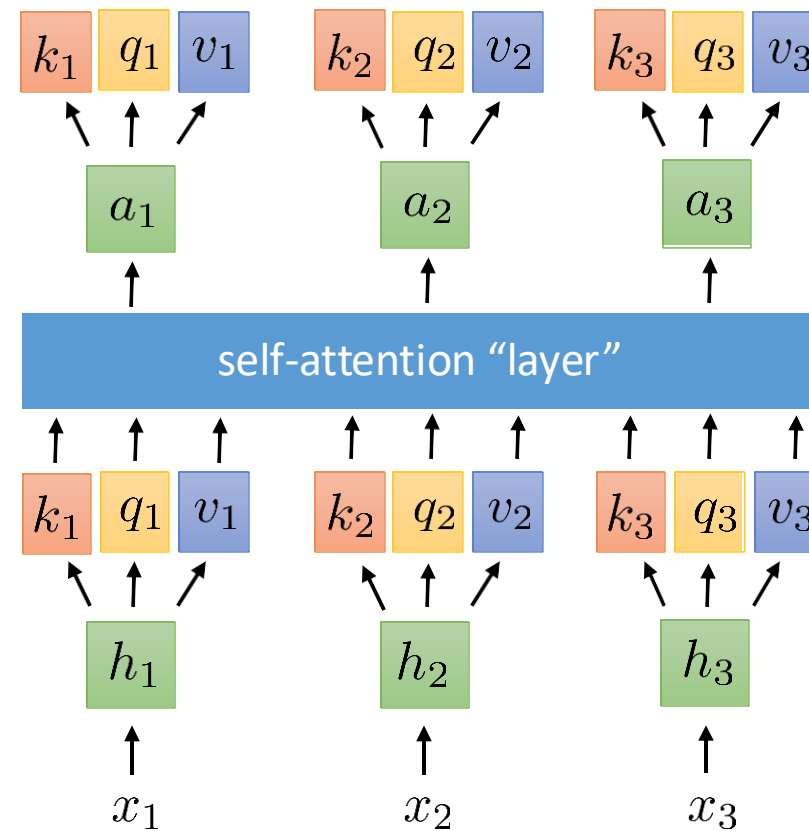
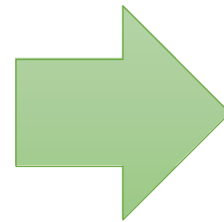
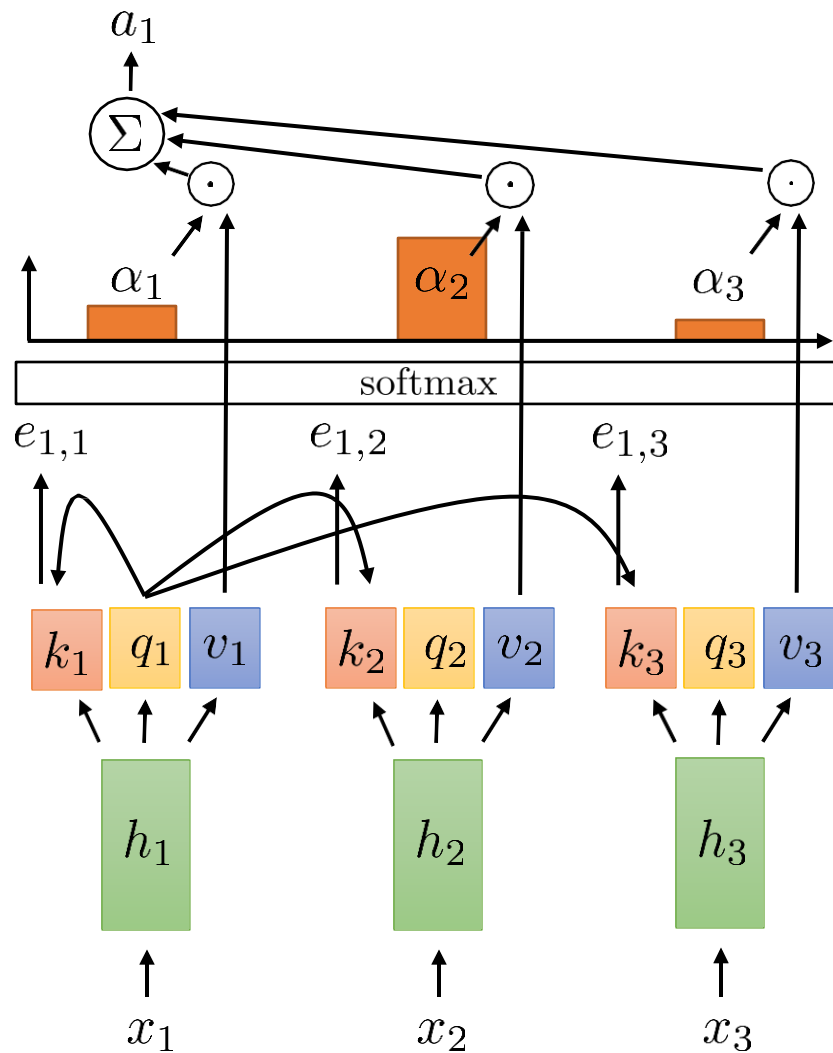
Self-Attention



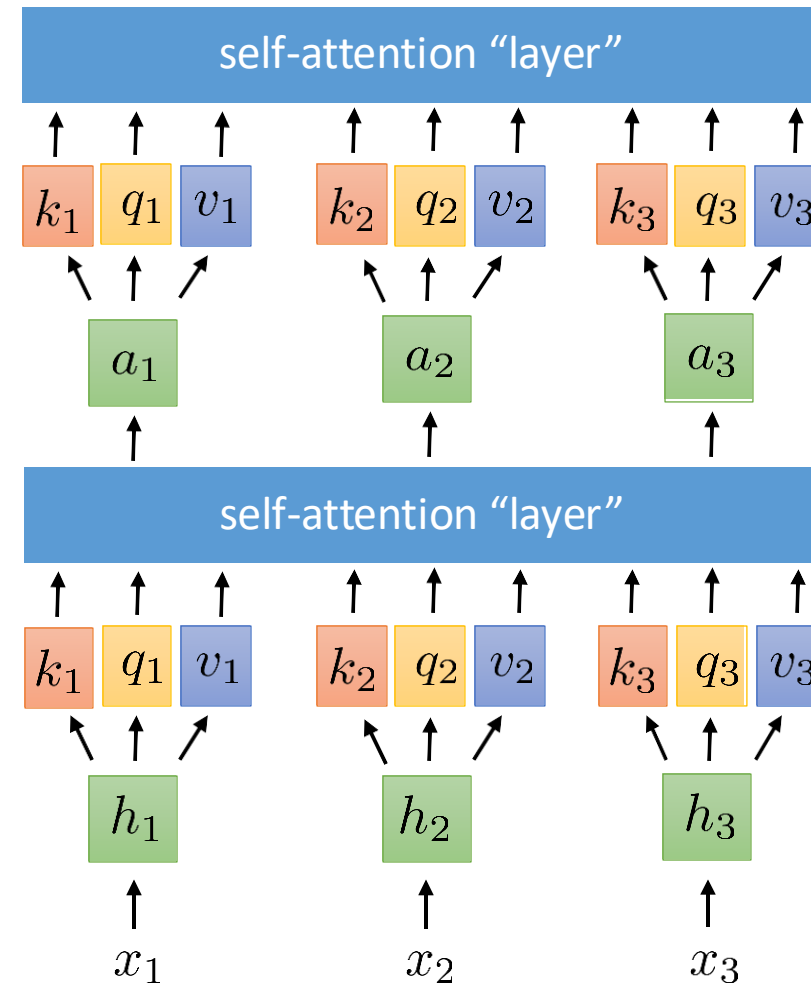
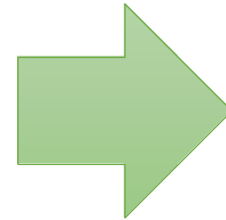
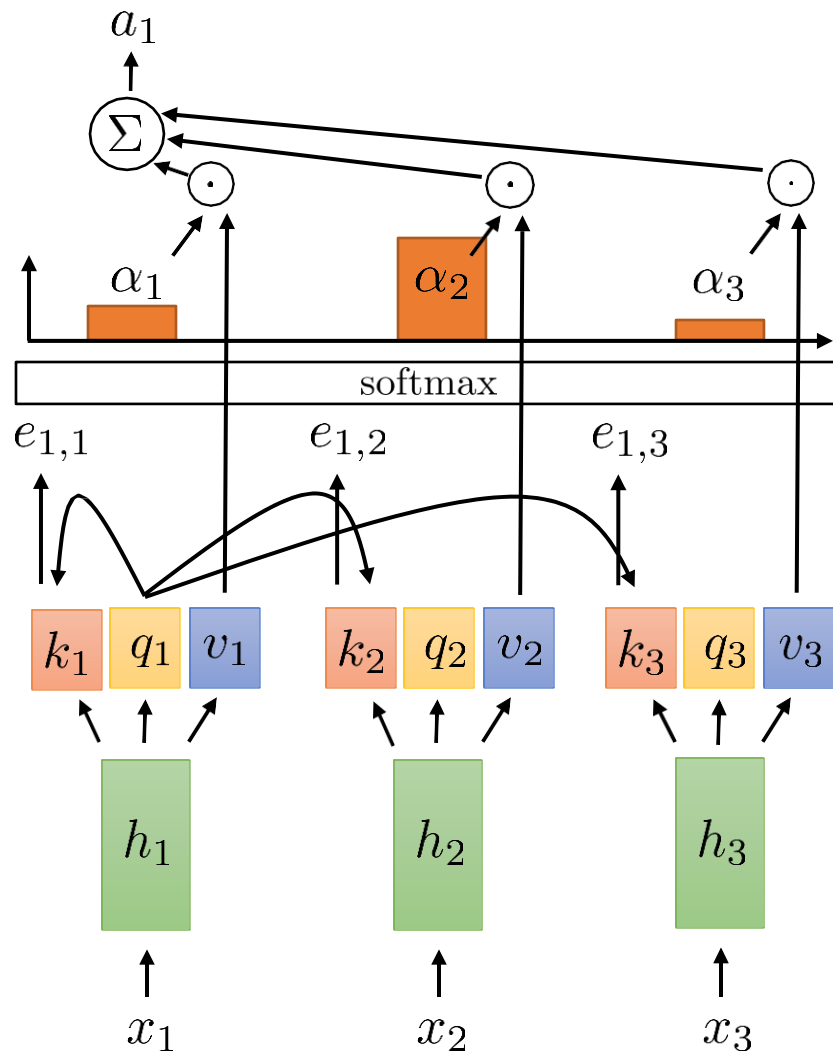
Self-Attention



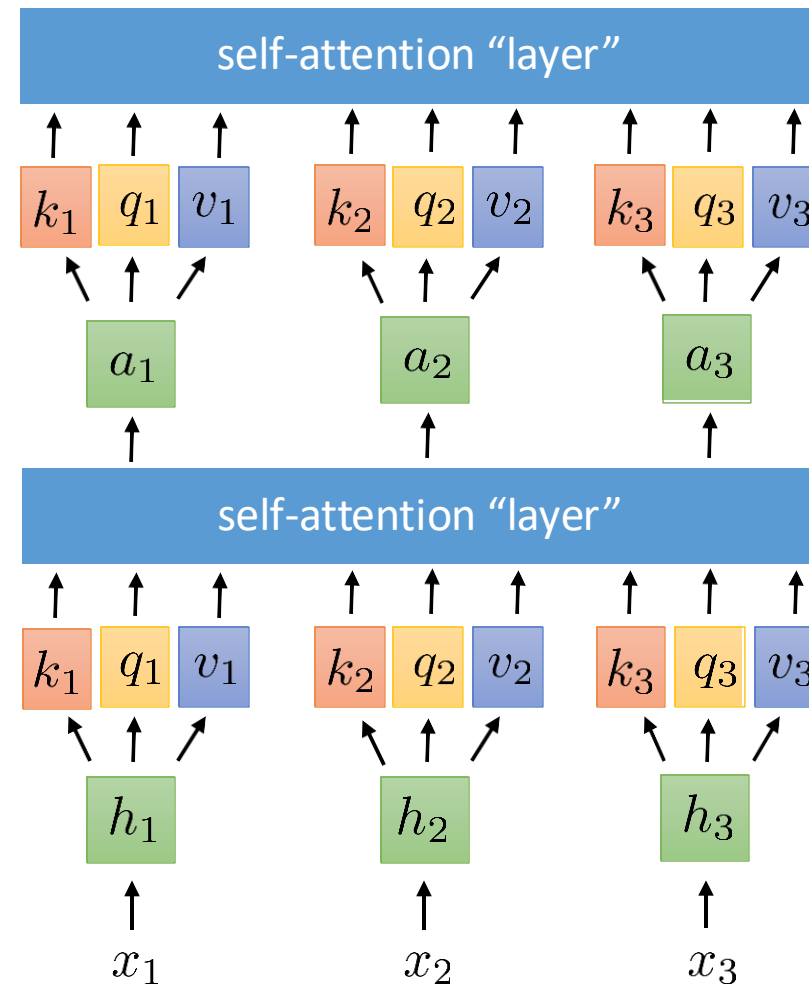
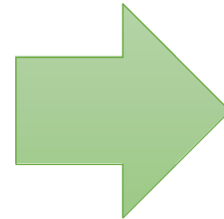
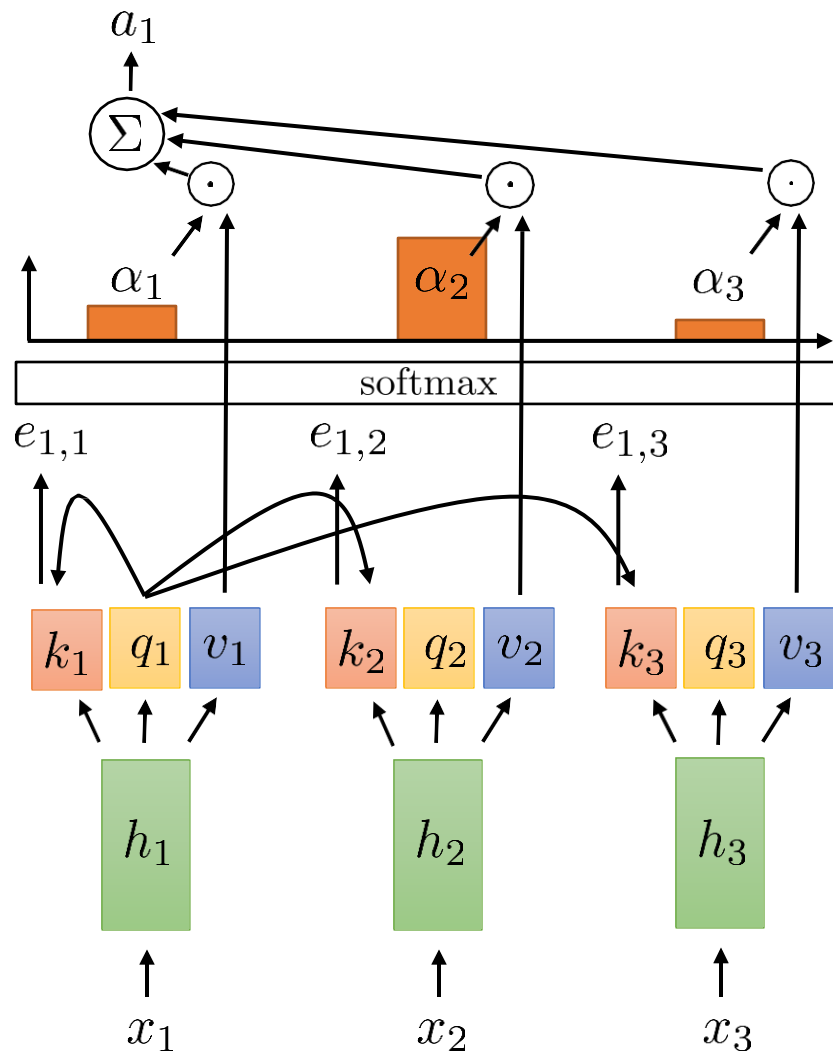
Self-Attention



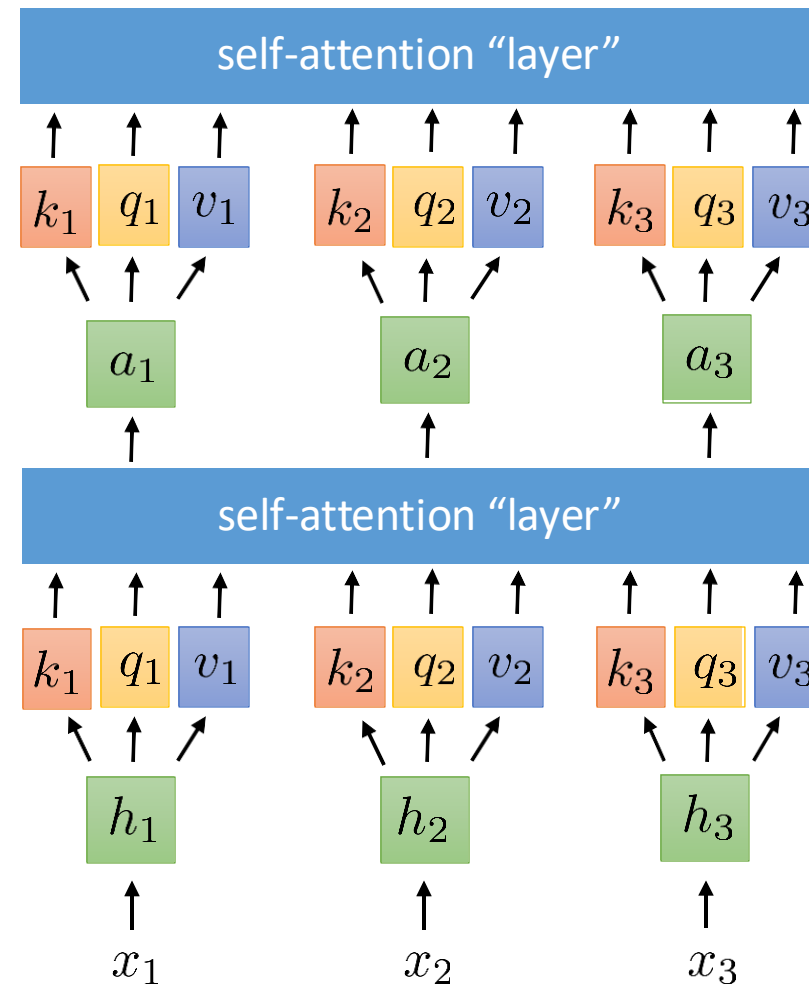
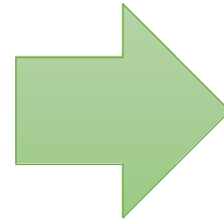
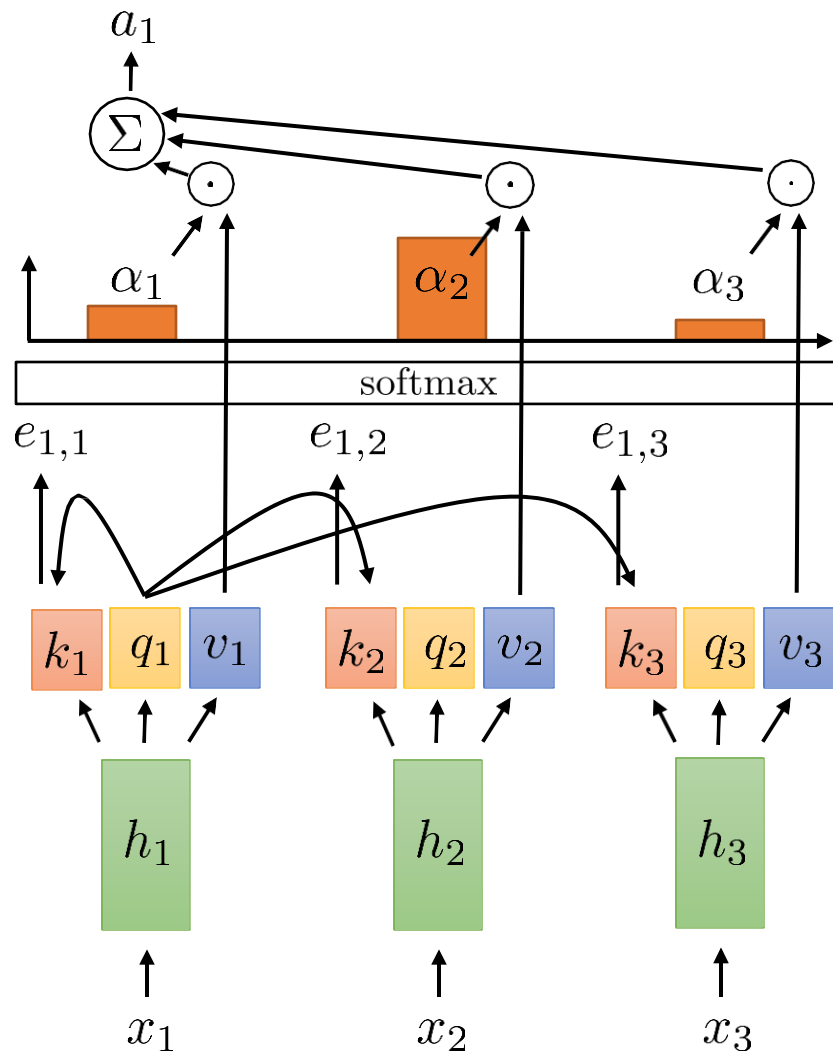
Self-Attention



Self-Attention



Self-Attention



From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called **Transformers**.
- To address a few key limitations, we need to add certain elements:
 1. Positional encoding addresses lack of sequence information
 2. Multi-headed attention allows querying multiple positions at each layer
 3. Adding nonlinearities so far, each successive layer is *linear* in the previous one
 4. Masked decoding how to prevent attention lookups into the future?

From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called **Transformers**.
- To address a few key limitations, we need to add certain elements:

1. Positional encoding

addresses lack of sequence information

2. Multi-headed attention

allows querying multiple positions at each layer

3. Adding nonlinearities

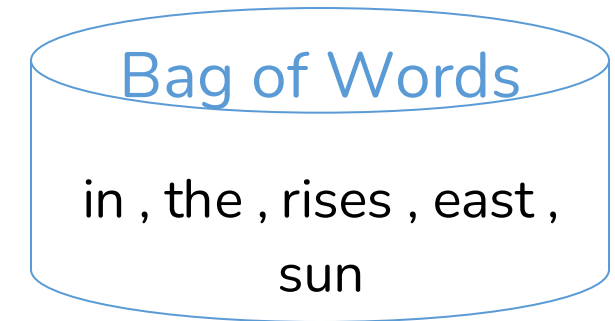
so far, each successive layer is *linear* in the previous one

4. Masked decoding

how to prevent attention lookups into the future?

Positional Encoding - Motivation

- **Problem** : Self-attention processes all the elements of a sequence in parallel without any regard for their order.
 - Example : the sun rises in the east
 - Permuted version : rises in the sun the east
the east rises in the sun
 - Self-attention is permutation invariant.
 - In natural language, it is important to take into account the order of words in a sentence.
- **Solution** : Explicitly add positional information to indicate where a word appears in a sequence

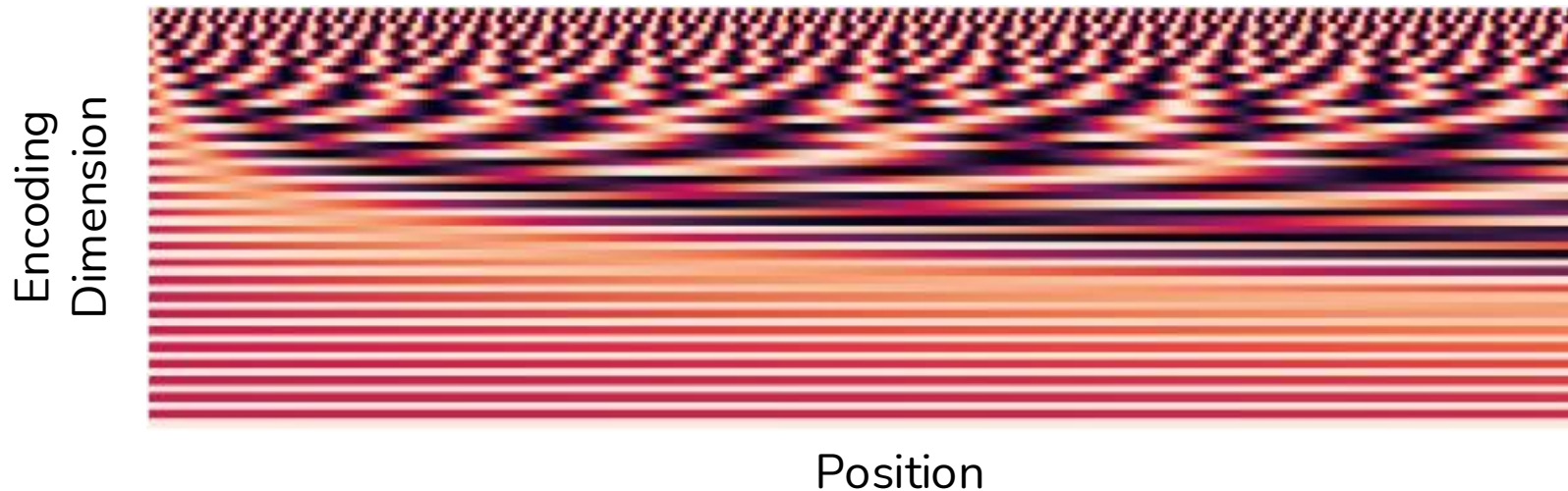


Sinusoidal Positional Encoding

- Helps it determine the position of each word (absolute positional information), or the distance between different words in the sequence (relative positional information)
- The frequency decreases along the encoding dimension.

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

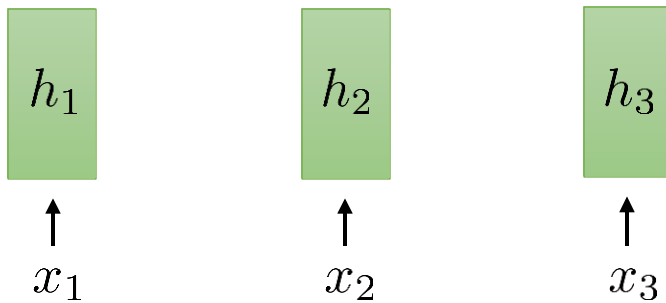


From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called transformers.
- To address a few key limitations, we need to add certain elements:
 1. Positional encoding addresses lack of sequence information
 2. Multi-headed attention allows querying multiple positions at each layer
 3. Adding nonlinearities so far, each successive layer is *linear* in the previous one
 4. Masked decoding how to prevent attention lookups into the future?

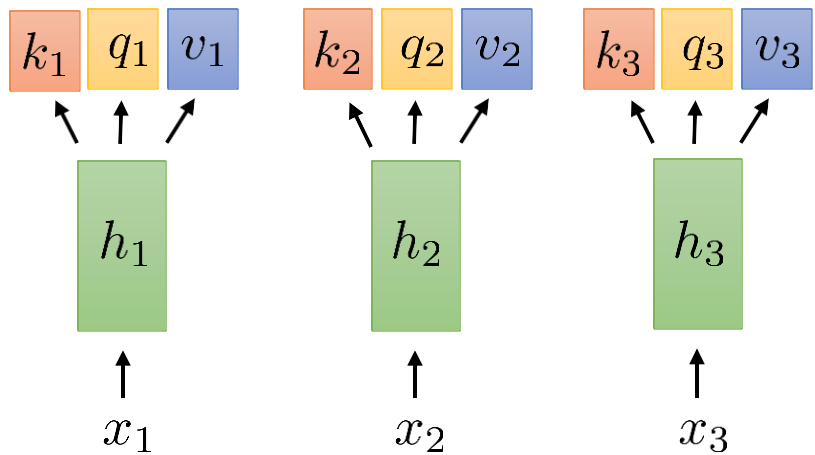
Multi-Head Attention

Given that we're fully depending on attention now, it could be beneficial to include more than one time step.



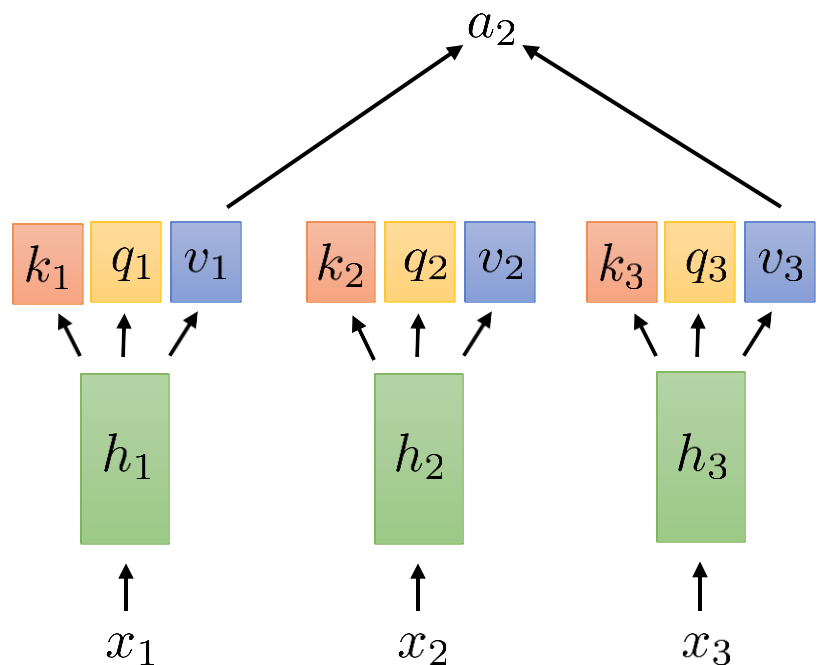
Multi-Head Attention

Given that we're fully depending on attention now, it could be beneficial to include more than one time step.



Multi-Head Attention

Given that we're fully depending on attention now, it could be beneficial to include more than one time step.

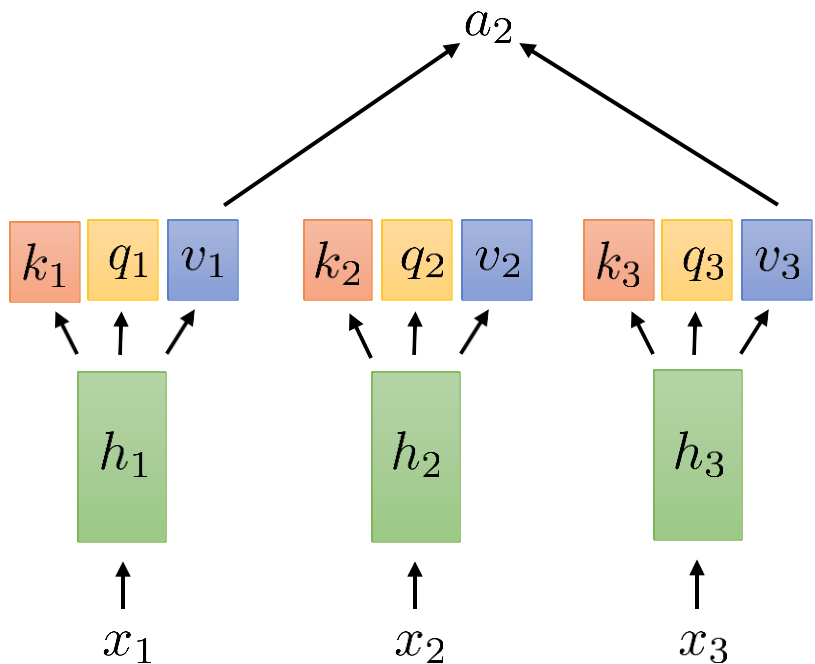


$$a_l = \sum_t \alpha_{l,t} v_t$$

Due to the softmax function, this will be heavily influenced by a single value.

Multi-Head Attention

Given that we're fully depending on attention now, it could be beneficial to include more than one time step.



$$a_l = \sum_t \alpha_{l,t} v_t$$

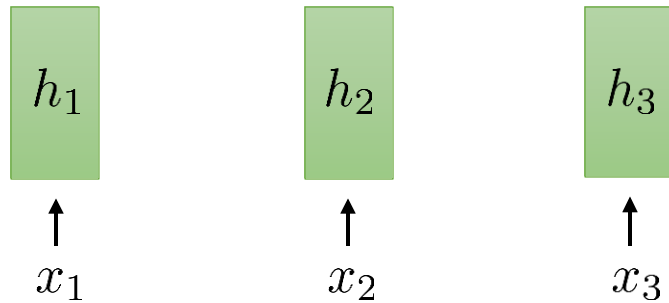
Due to the softmax function, this will be heavily influenced by a single value.

$$e_{l,t} = q_l \cdot k_t$$

It's challenging to clearly specify that you want two distinct elements, like the subject and object in a sentence.

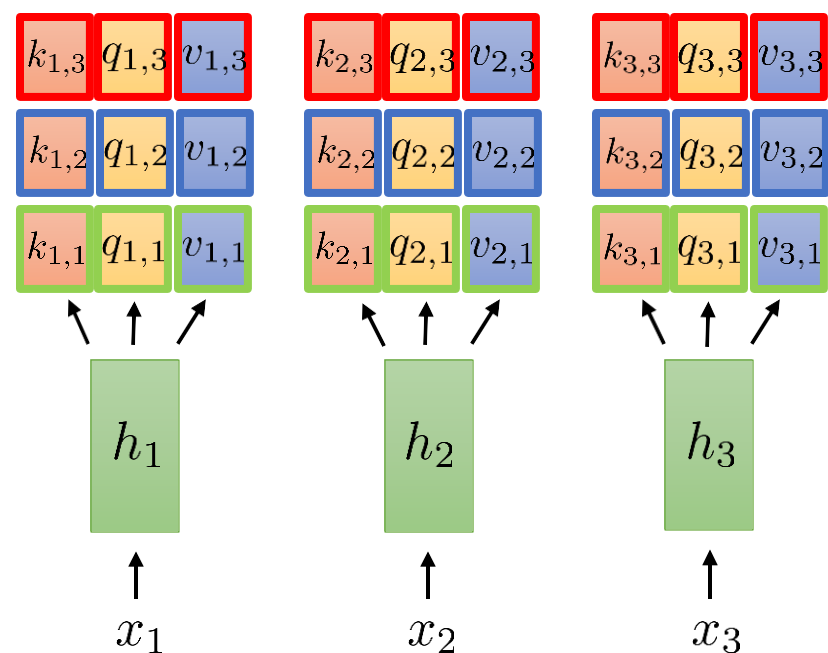
Multi-Head Attention

Solution: Use multiple keys, queries, and values for each time step



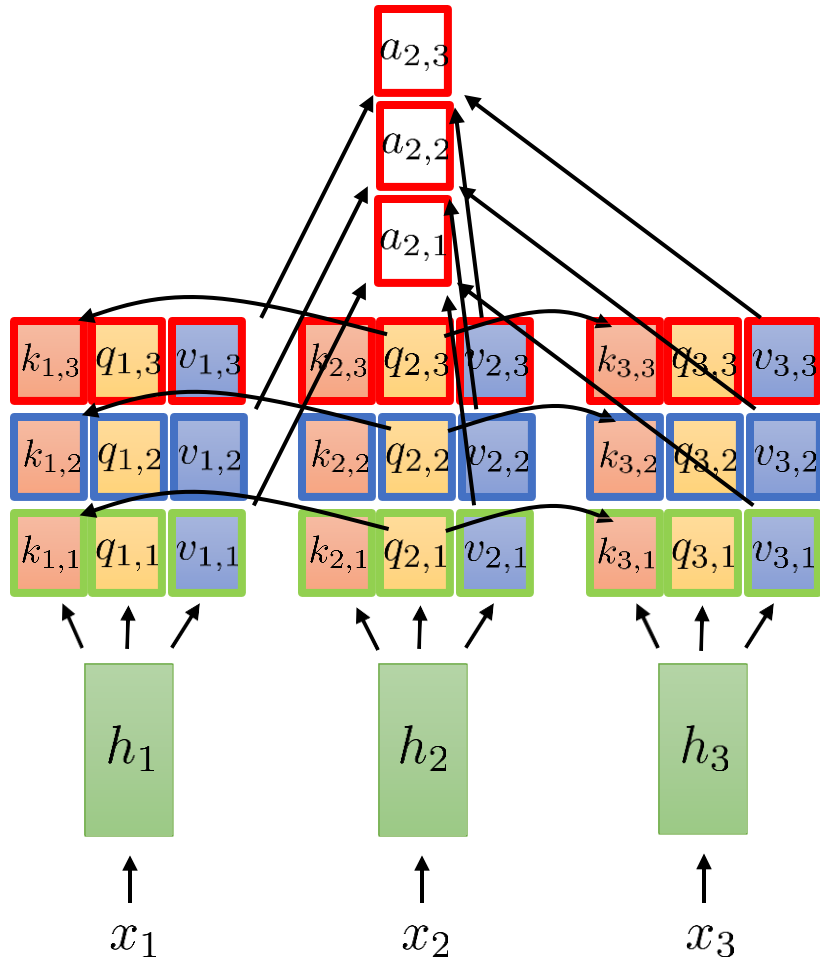
Multi-Head Attention

Solution: Use multiple keys, queries, and values for each time step



Multi-Head Attention

Solution: Use multiple keys, queries, and values for each time step



full attention vector formed by concatenation:

$$a_2 = \begin{bmatrix} a_{2,1} \\ a_{2,2} \\ a_{2,3} \end{bmatrix}$$

compute weights **independently** for each head

$$e_{l,t,i} = q_{l,i} \cdot k_{l,i}$$

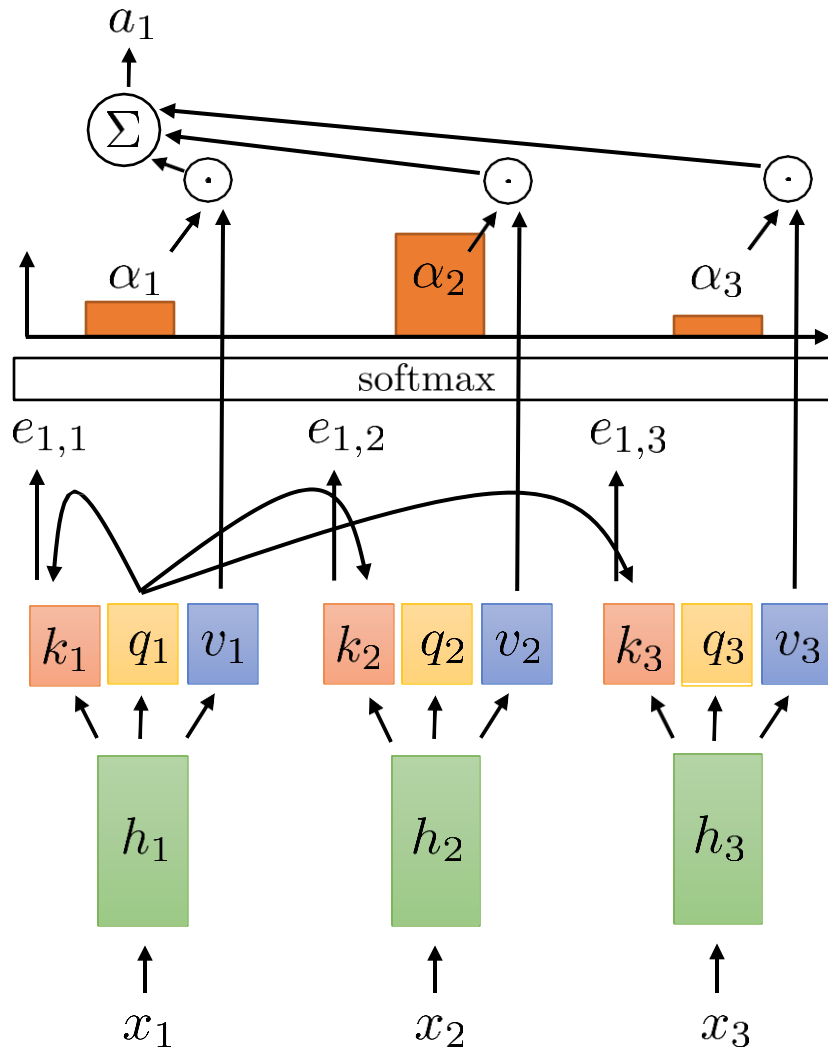
$$\alpha_{l,t,i} = \exp(e_{l,t,i}) / \sum_{t'} \exp(e_{l,t',i})$$

$$a_{l,i} = \sum_t \alpha_{l,t,i} v_{t,i}$$

From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called transformers.
- To address a few key limitations, we need to add certain elements:
 1. Positional encoding addresses lack of sequence information
 2. Multi-headed attention allows querying multiple positions at each layer
 3. Adding nonlinearities so far, each successive layer is *linear* in the previous one
 4. Masked decoding how to prevent attention lookups into the future?

Self-Attention Is “Linear”



$$k_t = W_k h_t \quad q_t = W_q h_t \quad v_t = W_v h_t$$

$$\alpha_{l,t} = \exp(e_{l,t}) / \sum_{t'} \exp(e_{l,t'})$$

$$e_{l,t} = q_l \cdot k_t$$

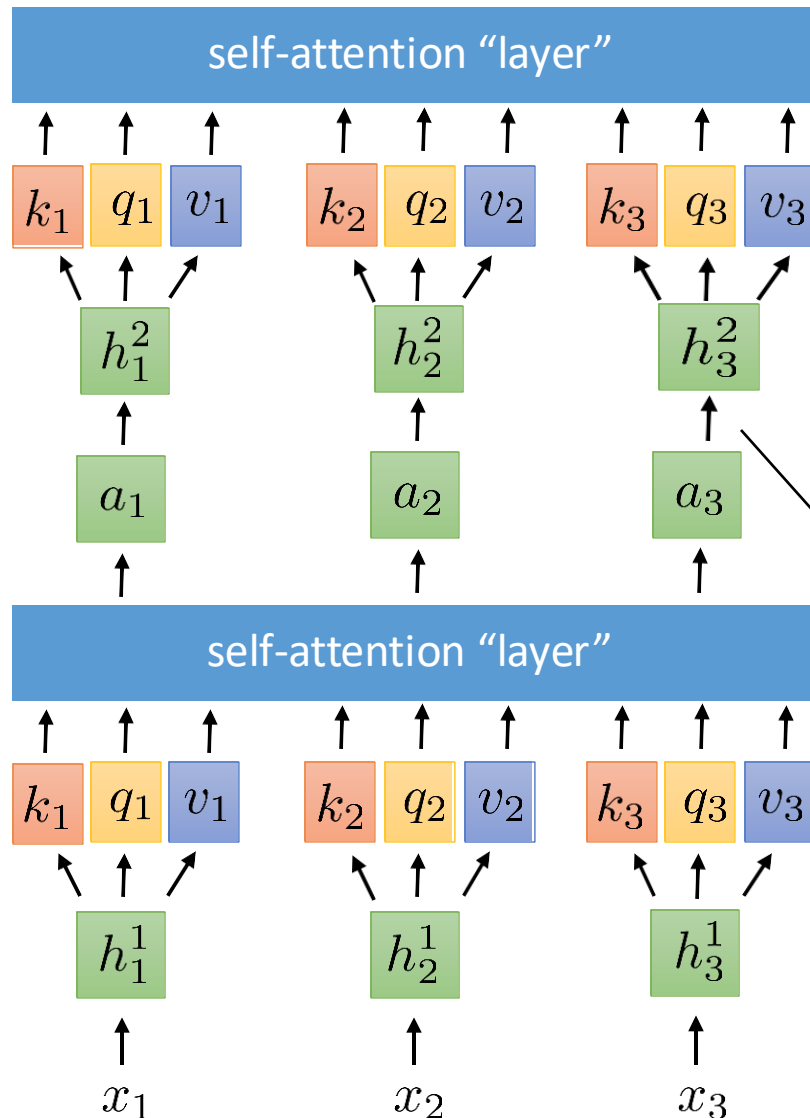
$$a_l = \sum_t \alpha_{l,t} v_t = \sum_t \alpha_{l,t} W_v h_t = W_v \sum_t \alpha_{l,t} h_t$$

linear transformation

non-linear weights

Problem: Every self-attention layer is a linear transformation of the previous layer with non-linear weights.

Position-wise Feed-Forward Networks



- **Solution :** Make the model more expressive is by alternating use of self-attention and non-linearity.
- Non-linearity is incorporated by means of a feed-forward network which consists of two linear transformations with a ReLU activation in between.

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

- The same non-linearity is utilized across various positions but they differ from layer to layer.

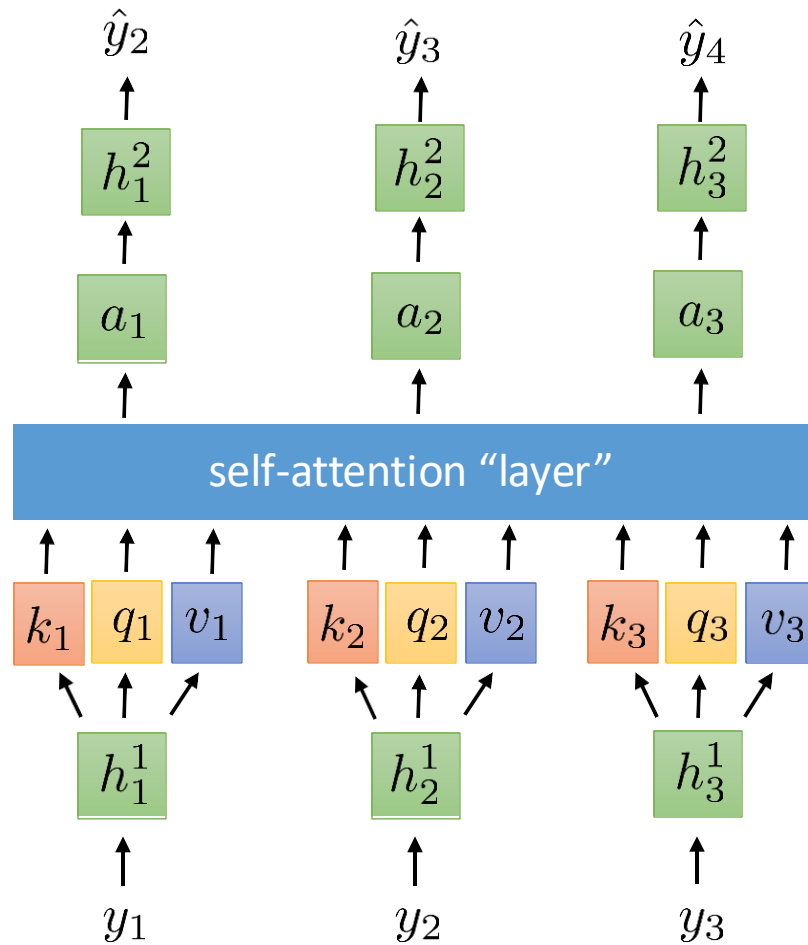
From Self-Attention to Transformers

- We will talk about a class of models for processing sequences that does not use recurrent connections but instead relies entirely on attention and will build up towards a class of models called transformers.
- To address a few key limitations, we need to add certain elements:
 1. Positional encoding addresses lack of sequence information
 2. Multi-headed attention allows querying multiple positions at each layer
 3. Adding nonlinearities so far, each successive layer is *linear* in the previous one
 4. Masked decoding how to prevent attention lookups into the future?

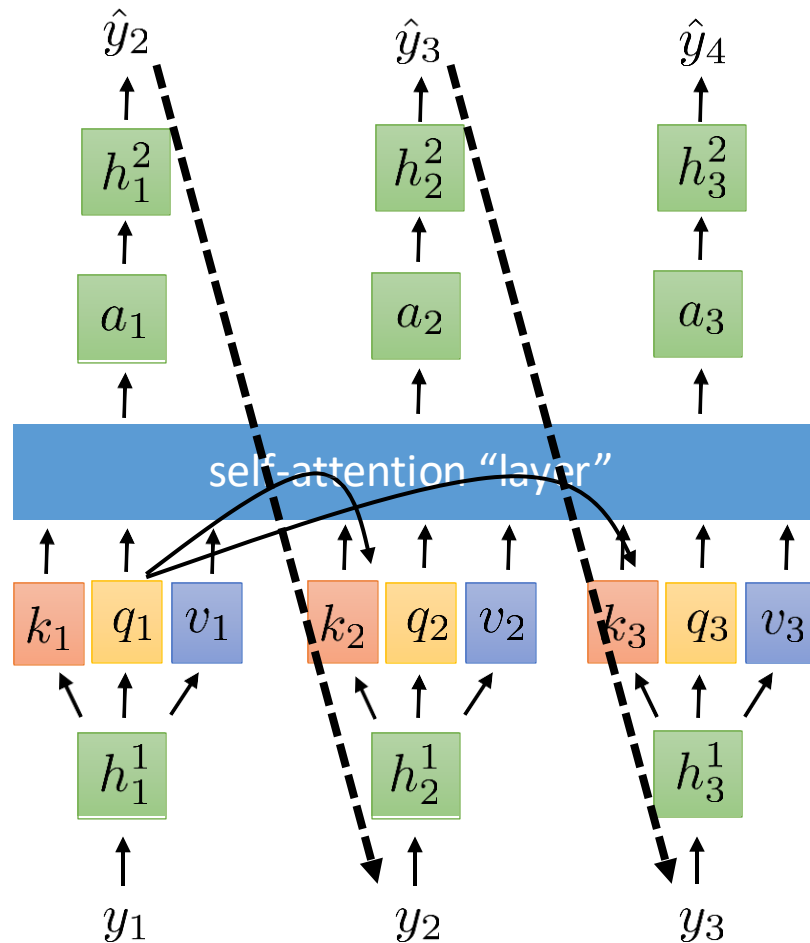
Self-attention can see the future!

A **crude** self-attention “language model”:

In practice, there would be several alternating self-attention layers and position-wise feedforward networks



Self-attention can see the future!



A **crude** self-attention “language model”:

In practice, there would be several alternating self-attention layers and position-wise feedforward networks

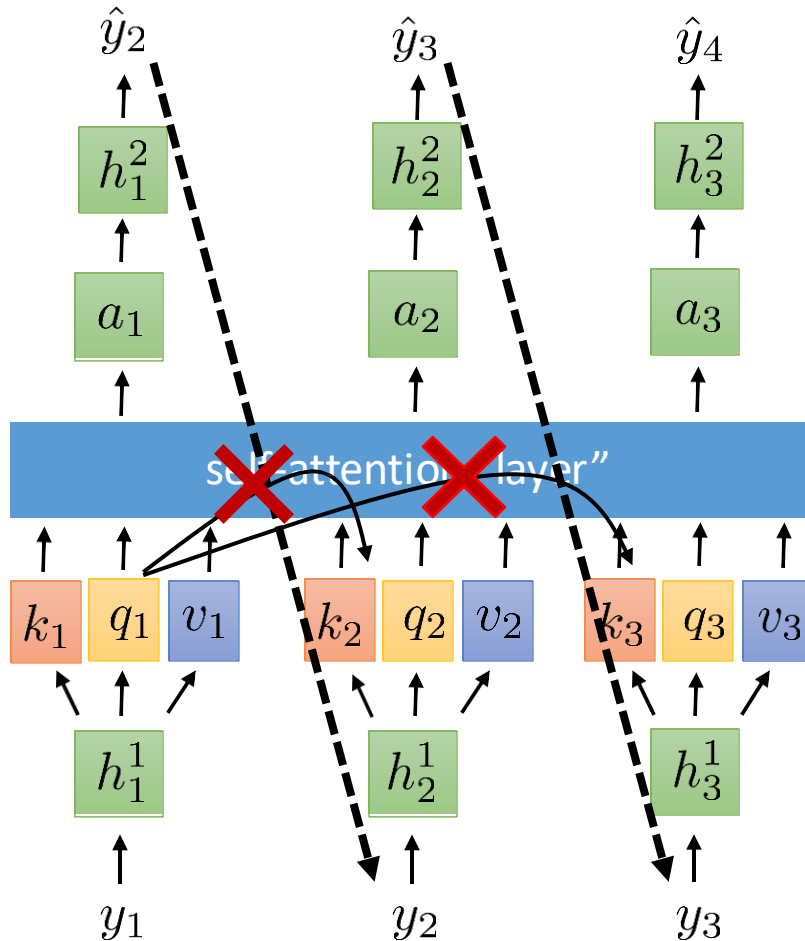
Big problem: self-attention at step 1 can look at the value at steps 2 & 3, which is based on the **inputs** at steps 2 & 3

At test time (when decoding), the **inputs** at steps 2 & 3 will be based on the output at step 1...

...which requires knowing the **input** at steps 2 & 3

Masked Attention

A **crude** self-attention “language model”:



At test time (when decoding), the **inputs** at steps 2 & 3 will be based on the output at step 1...

...which requires knowing the **input** at steps 2 & 3

Must allow self-attention into the **past**...

...but not into the **future**

Easy solution:

$$\cancel{e_{l,t} = q_l \cdot k_t}$$

$$e_{l,t} = \begin{cases} q_l \cdot k_t & \text{if } l \geq t \\ -\infty & \text{otherwise} \end{cases}$$

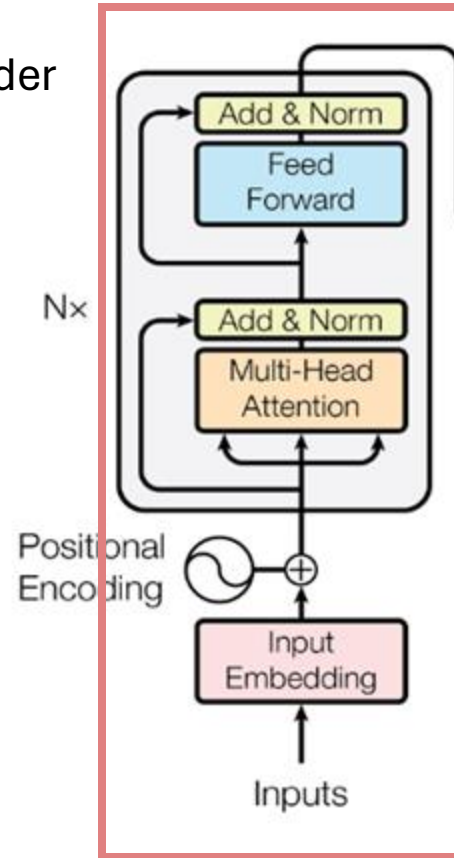
in practice:

just replace $\exp(e_{l,t})$ with 0 if $l < t$

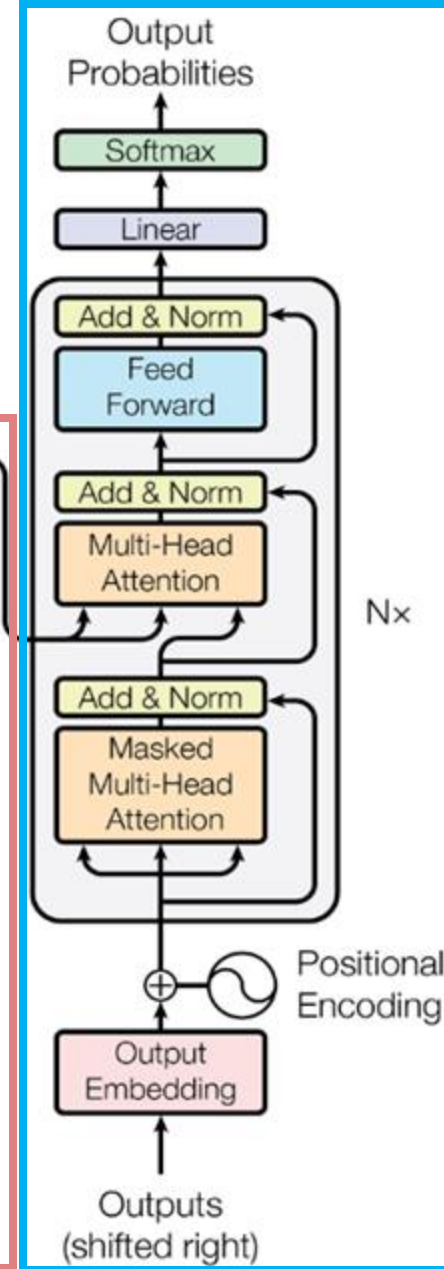
inside the softmax

Transformer Architecture

Encoder

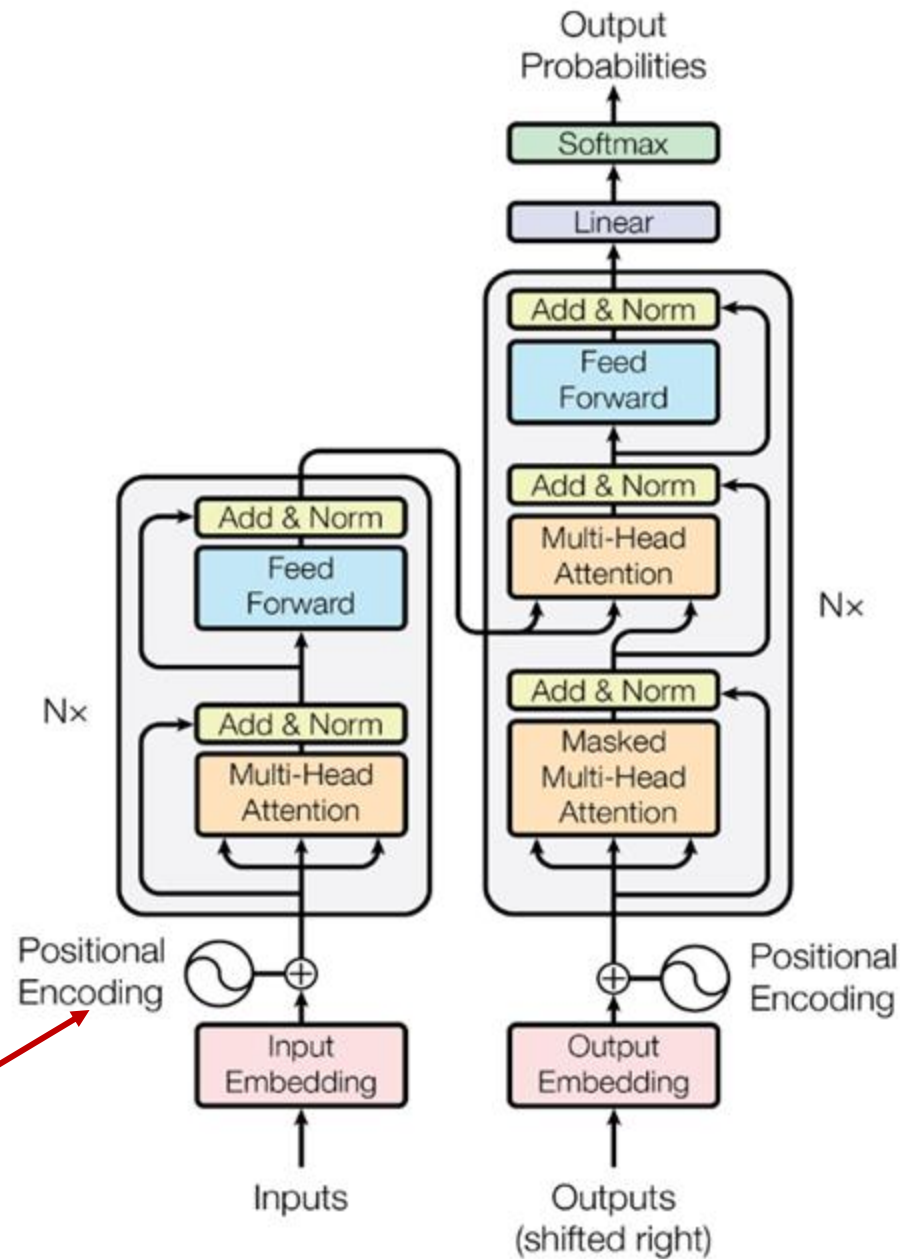


Decoder



Source of Image : Attention is all you need
(Vaswani et al., 2017)

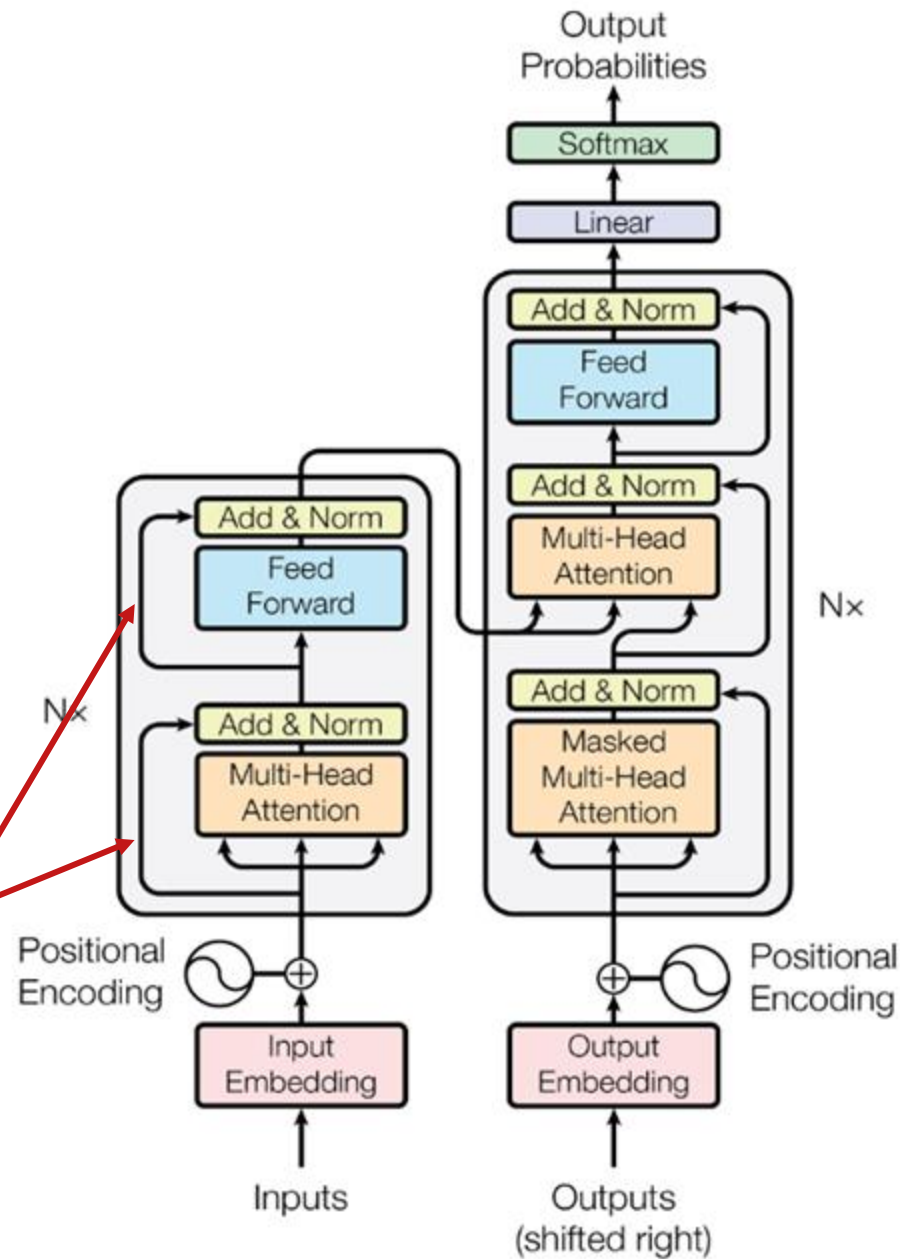
Transformer Architecture



Position embeddings are *added* to each word embedding. Otherwise, since we have no recurrence, our model is unaware of the position of a word in the sequence!

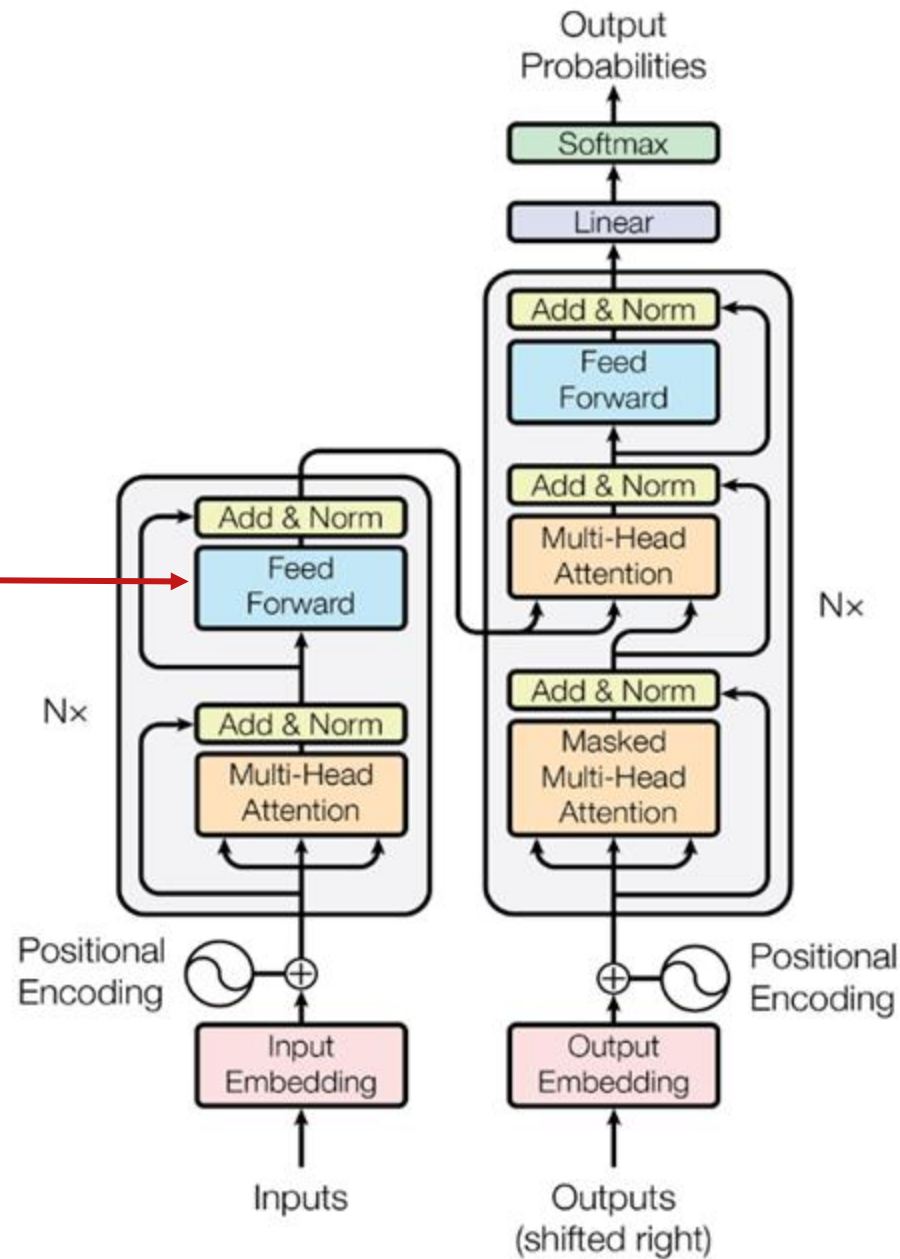
Transformer Architecture

Residual connections, which mean that we add the input to a particular block to its output, help improve gradient flow



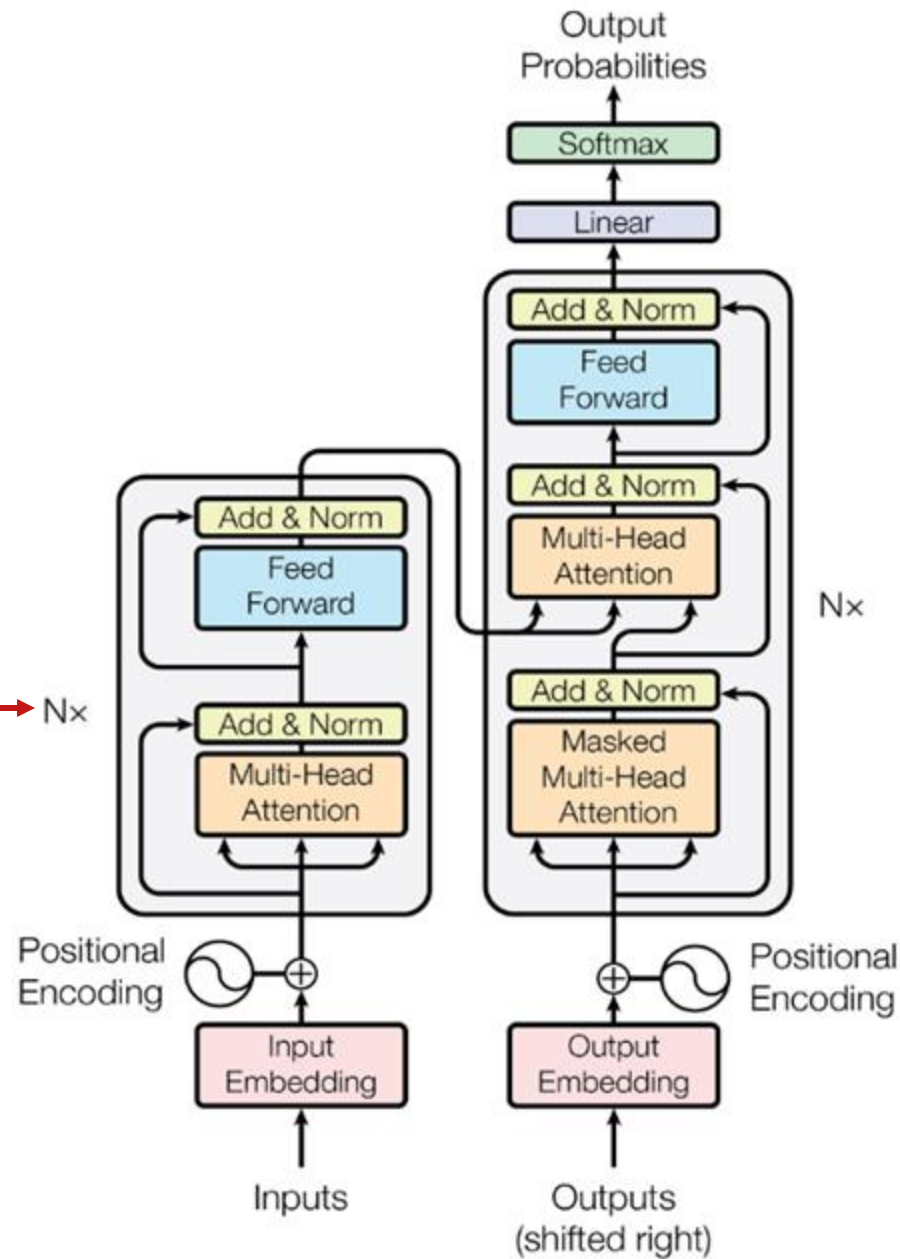
Transformer Architecture

A **feed-forward layer** on top of the attention- weighted averaged value vectors allows us to add more parameters / nonlinearity

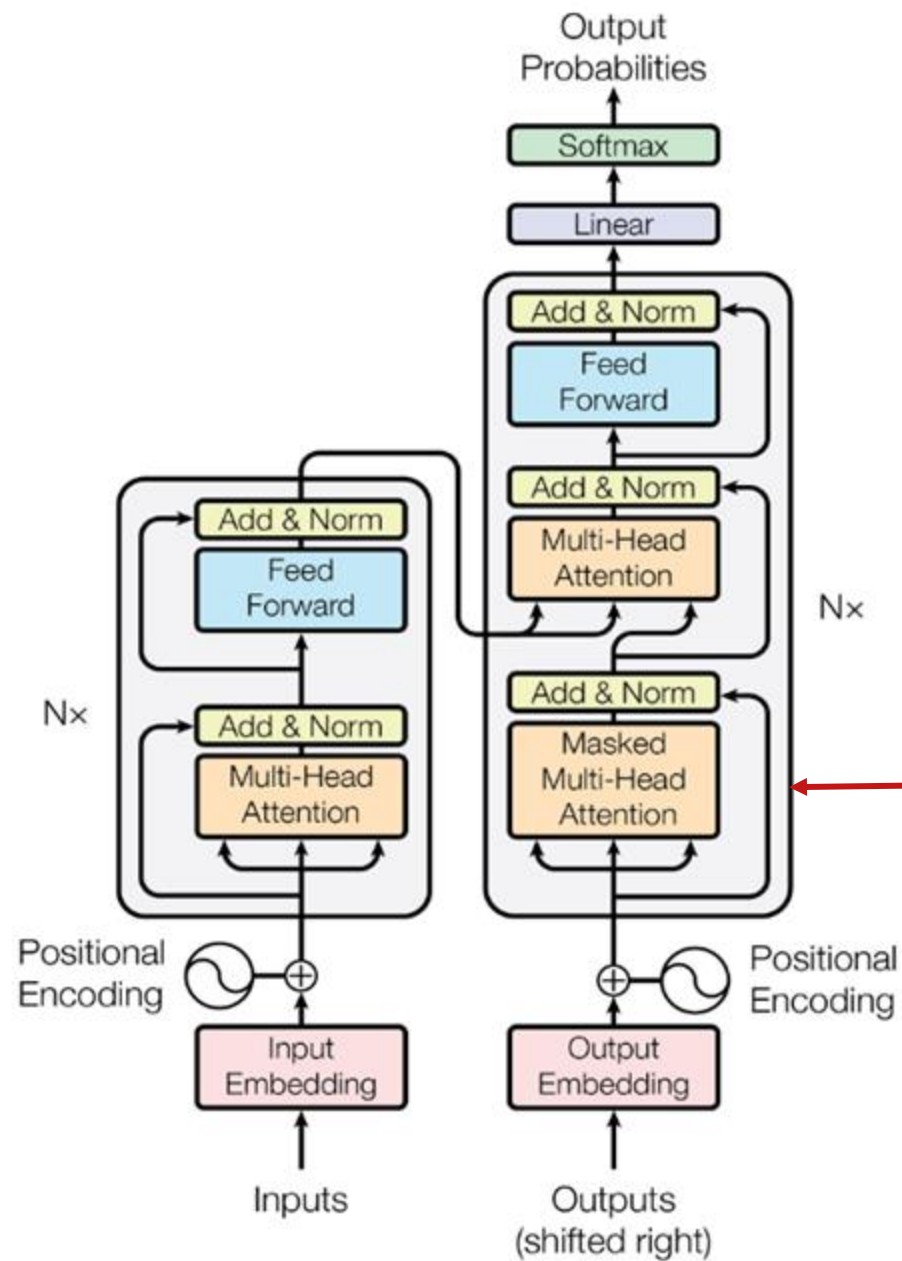


Transformer Architecture

We stack as many of these **Transformer blocks** on top of each other as we can (bigger models are generally better given enough data!)

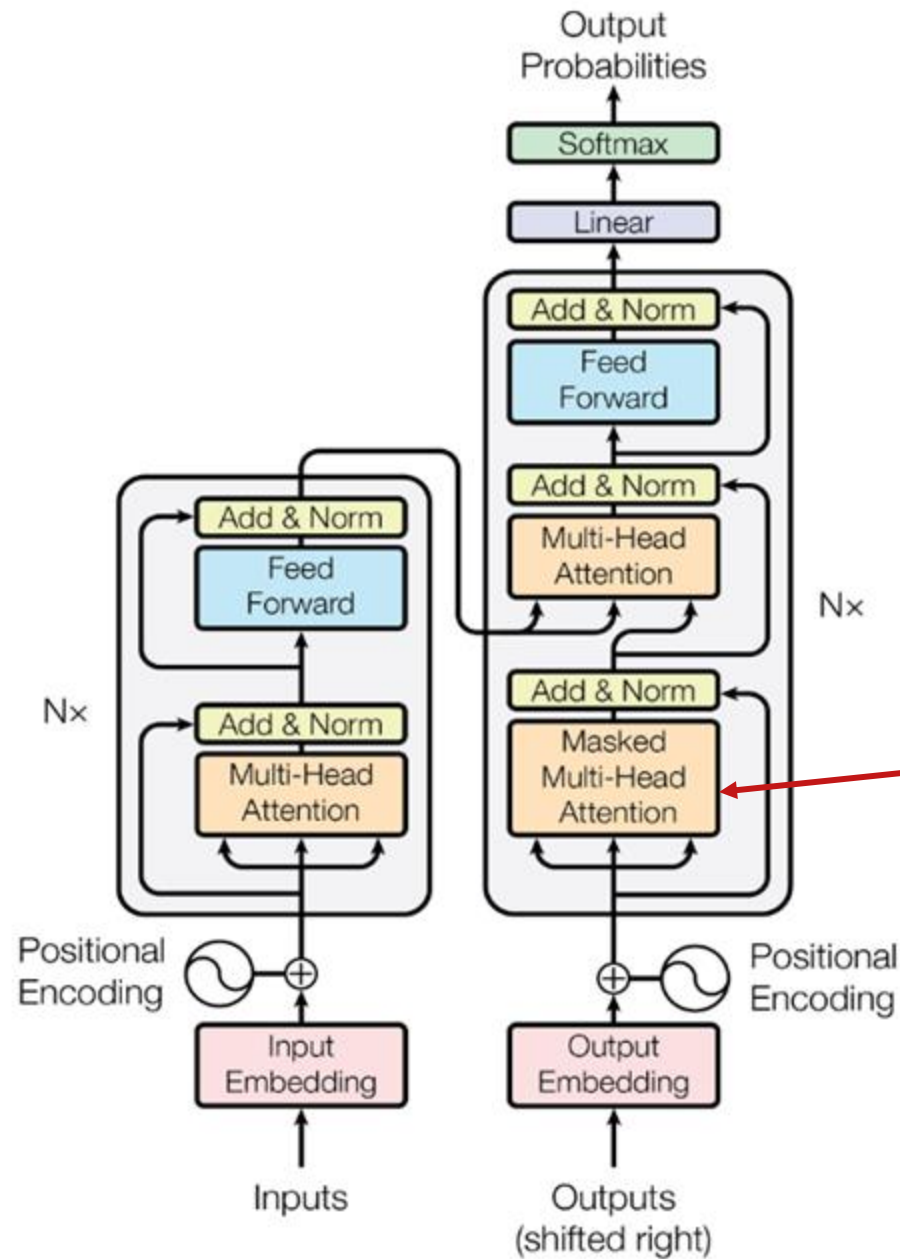


Transformer Architecture



Moving onto the decoder, which takes in English sequences that have been shifted to the right (e.g., *<START> schools opened their*)

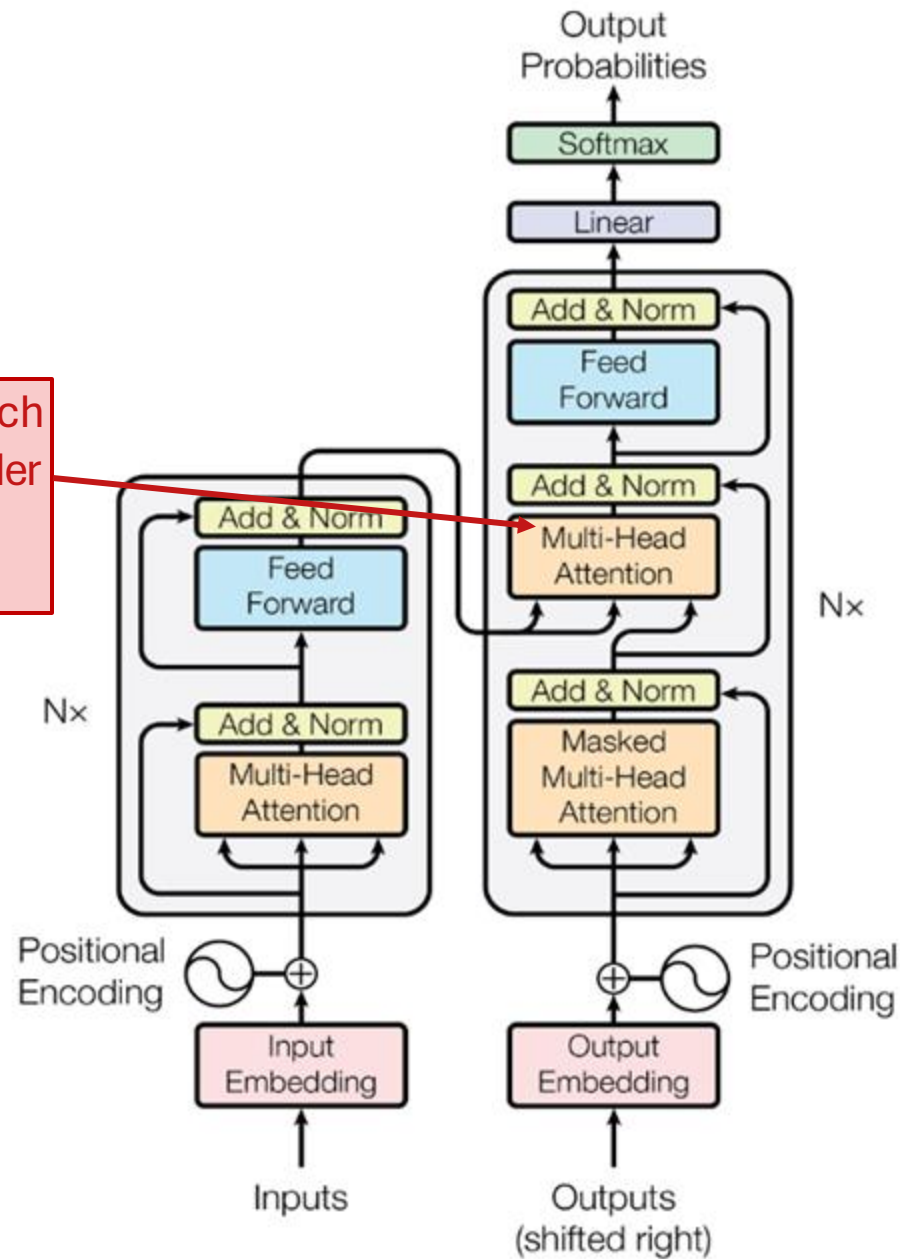
Transformer Architecture



We first have an instance of **masked self attention**. Since the decoder is responsible for predicting the English words, we need to apply masking as we saw before.

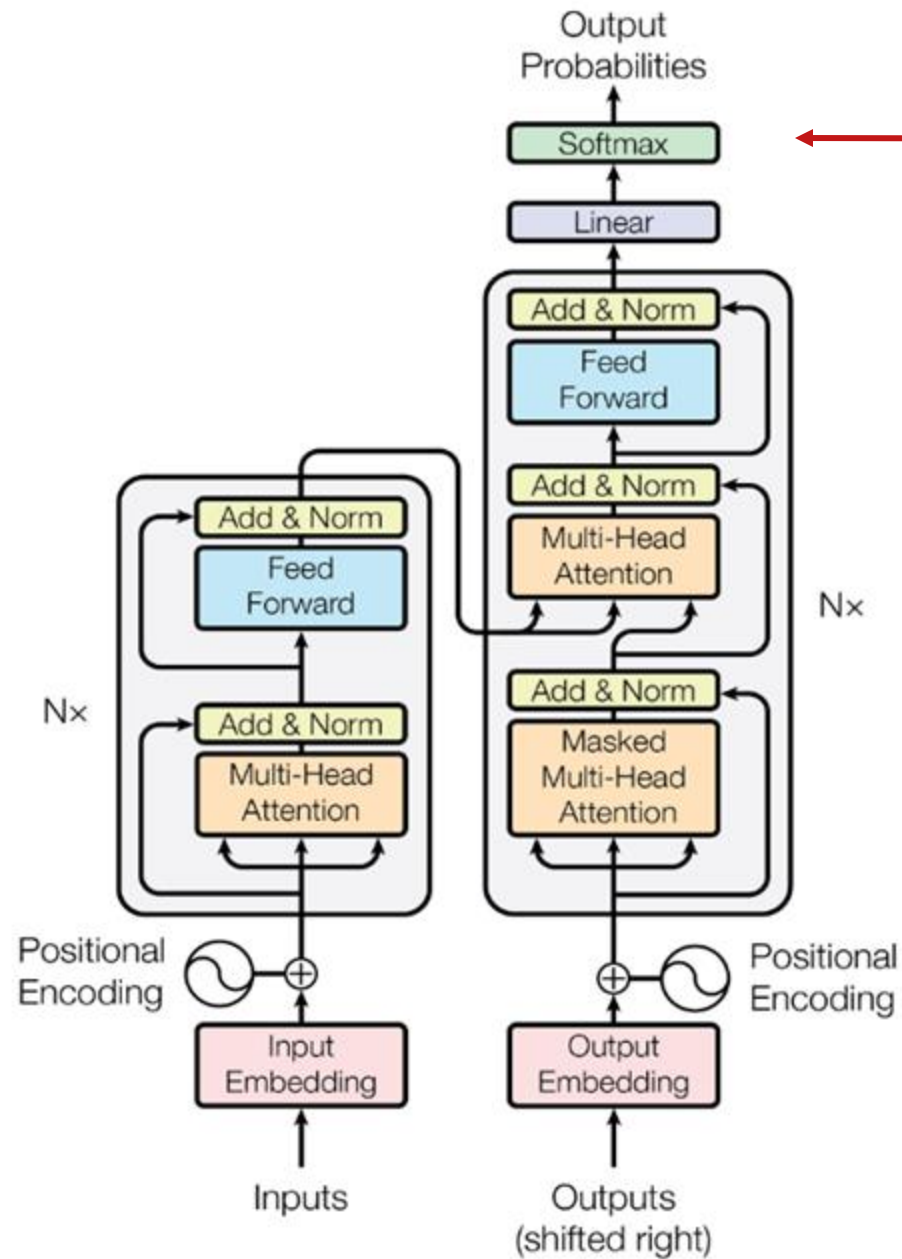
Transformer Architecture

Now, we have **cross attention**, which connects the decoder to the encoder by enabling it to attend over the encoder's final hidden states.



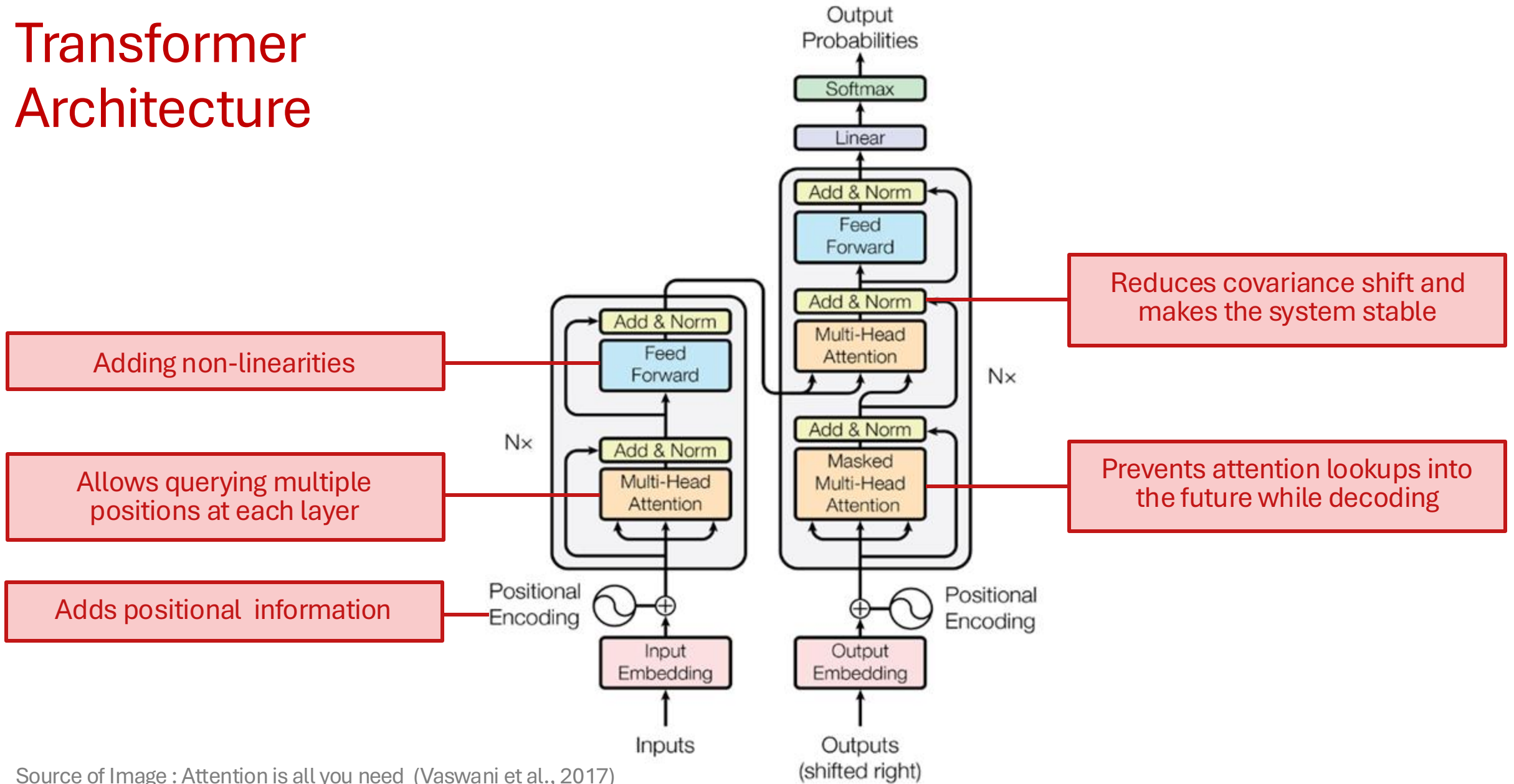
Source of Image : Attention is all you need
(Vaswani et al., 2017)

Transformer Architecture



After stacking a bunch of these decoder blocks, we finally have our familiar **softmax** layer to predict the next English word.

Transformer Architecture



Source of Image : Attention is all you need (Vaswani et al., 2017)

Layer normalization

- **Main idea:** Batch normalization is quite beneficial, but it's challenging to apply with sequence models. The varying lengths of sequences make it difficult to normalize across a batch. Sequences can be very long, which often results in smaller batch sizes.
- **Solution:** Layer normalization

$a_1, a_2, \dots, a_B$ ← d -dimensional vectors for each sample in batch

d -dim → $\mu = \frac{1}{B} \sum_{i=1}^B a_i$ $\sigma = \sqrt{\frac{1}{B} \sum_{i=1}^B (a_i - \mu)^2}$

1-dim → $\mu = \frac{1}{d} \sum_{i=1}^d a_j$ $\sigma = \sqrt{\frac{1}{d} \sum_{i=1}^d (a_j - \mu)^2}$ ← different *dimensions* of a

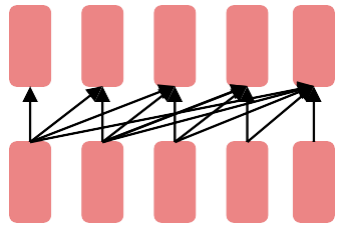
$\bar{a}_i = \frac{a_i - \mu}{\sigma} \gamma + \beta$ **Batch Norm**

$\bar{a} = \frac{a - \mu}{\sigma} \gamma + \beta$ **Layer Norm**

Pre-training Strategies

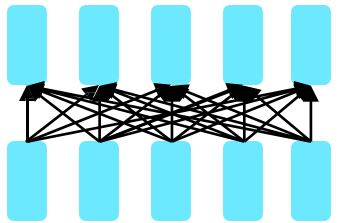
Pretraining for Three Types of Architectures

The neural architecture influences the type of pretraining, and natural use cases.



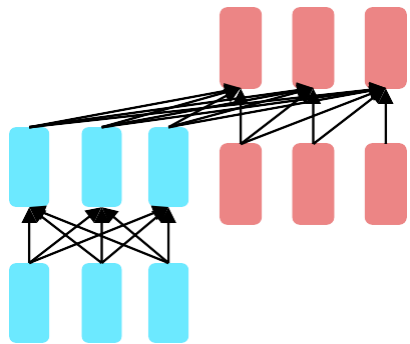
Decoders

- Language models! What we've seen so far.
- Nice to generate from; can't condition on future words



Encoders

- Gets bidirectional context – can condition on future!
- How do we pretrain them?



**Encoder-
Decoders**

- Good parts of decoders and encoders?
- What's the best way to pretrain them?

BERT: Bidirectional Encoder Representations from Transformers

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova

Google AI Language

`{jacobdevlin, mingweichang, kentonl, kristout}@google.com`

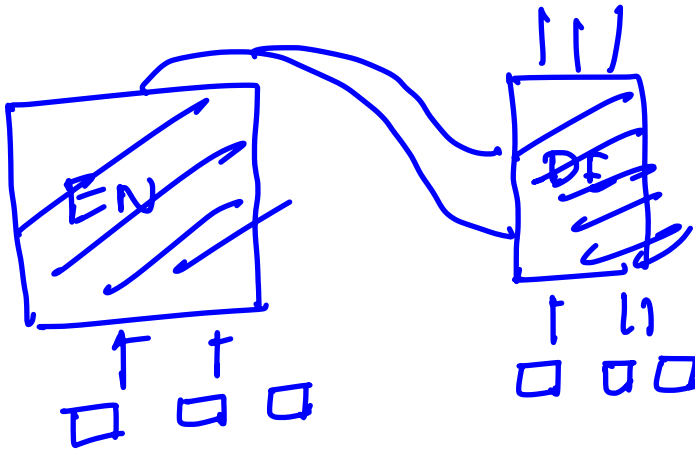
Slides are adopted from Jacob Devlin

Pre-trained word embedding - Word2vec
Pre-trained Model

BERT - Bidirectional
Encoder Representation of
Transformers

MLM
Masked

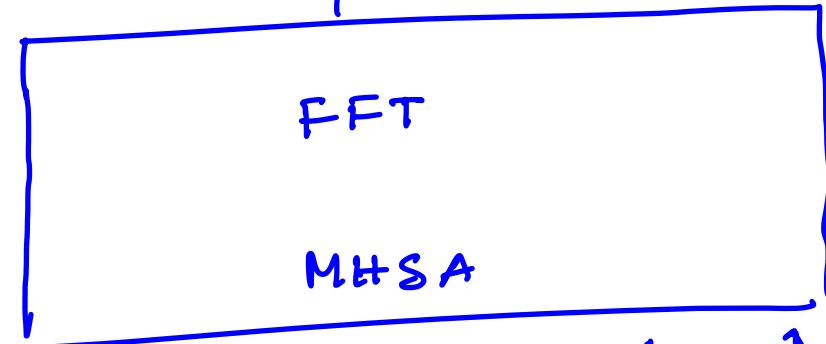
Language Models



Fine Tune
↑
Pre-train

a_1 a_2 a_3 a_4 a_5

↑
⋮

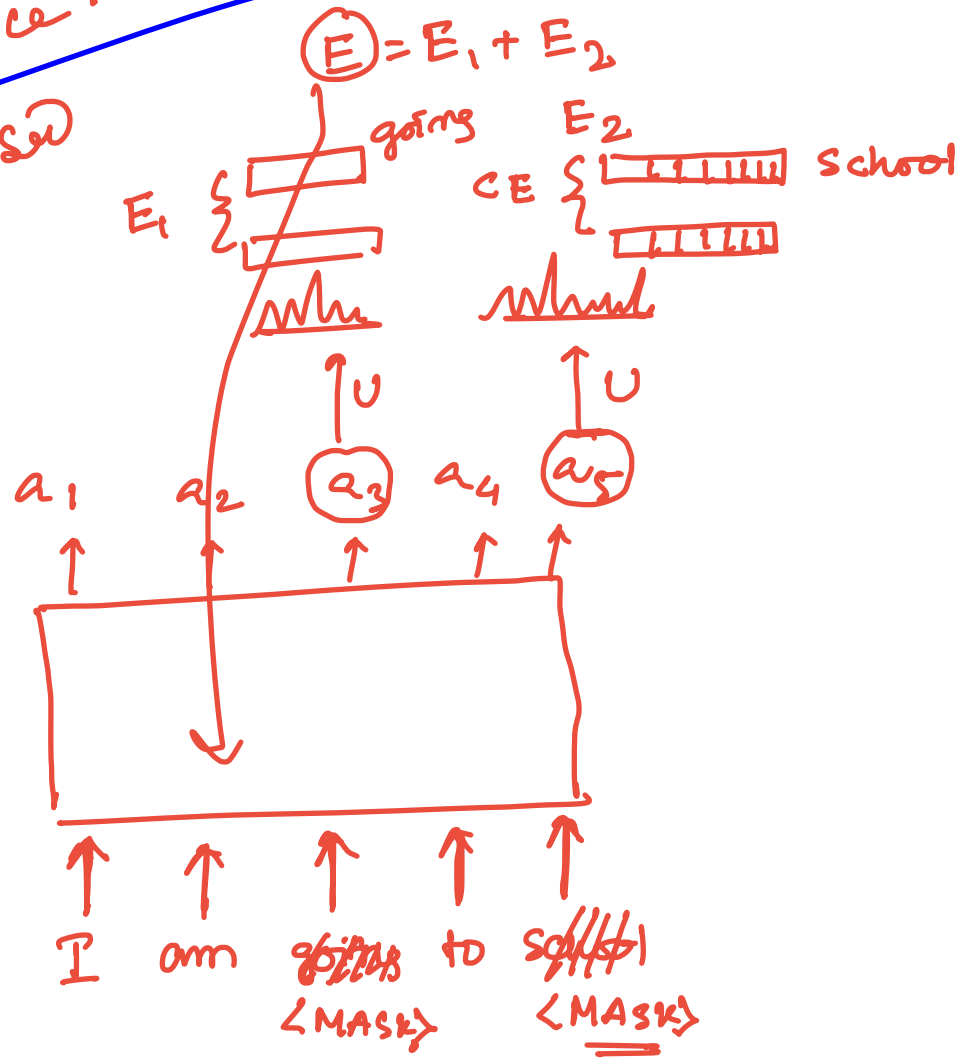


MLM

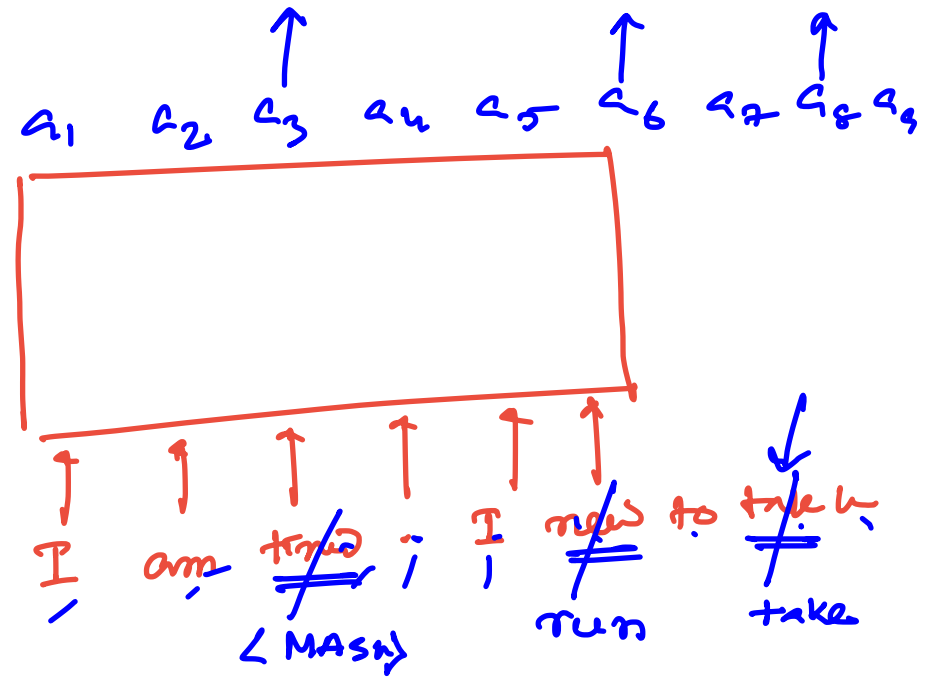
Masked token Prediction

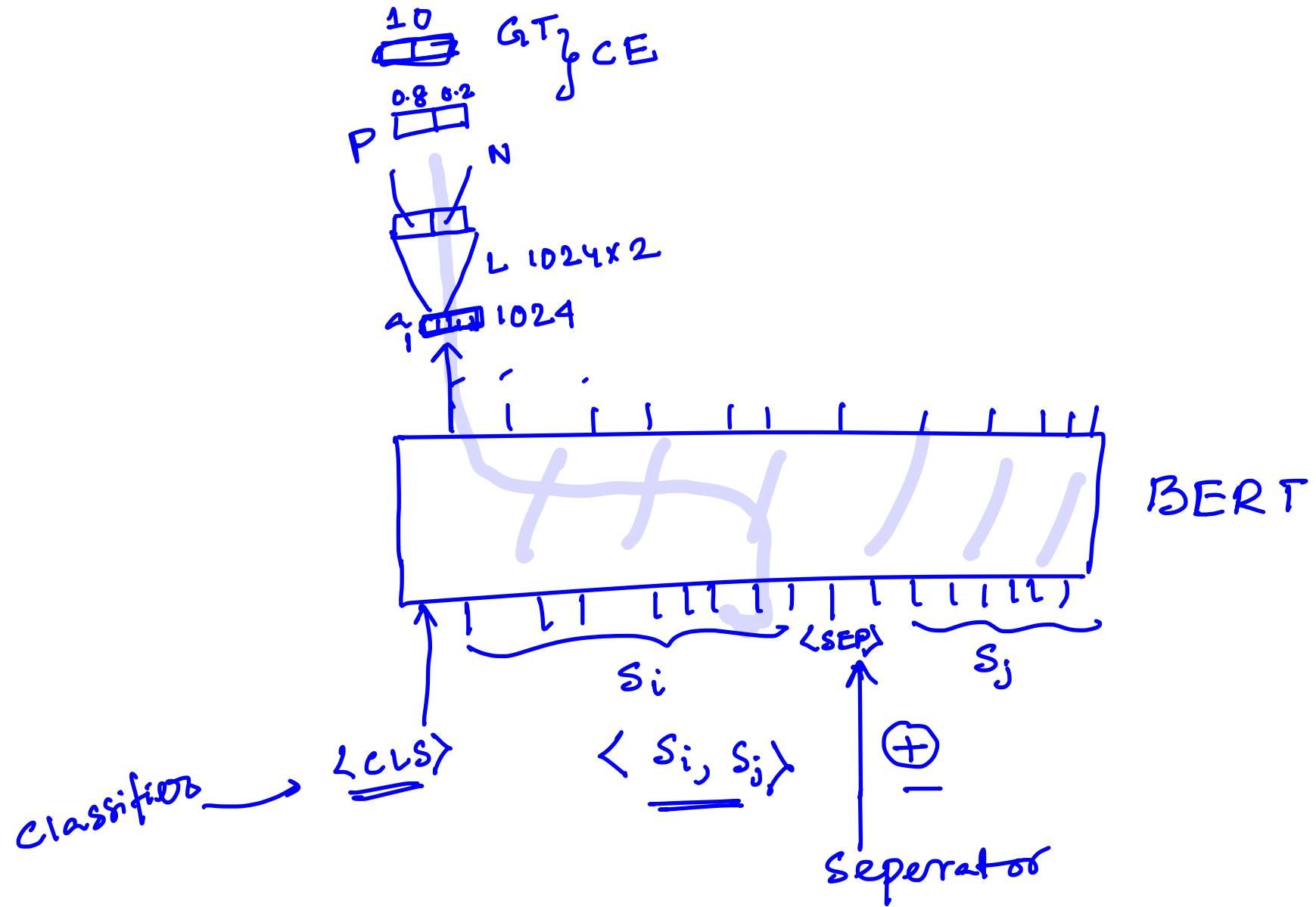
Next Sentence Prediction

BART

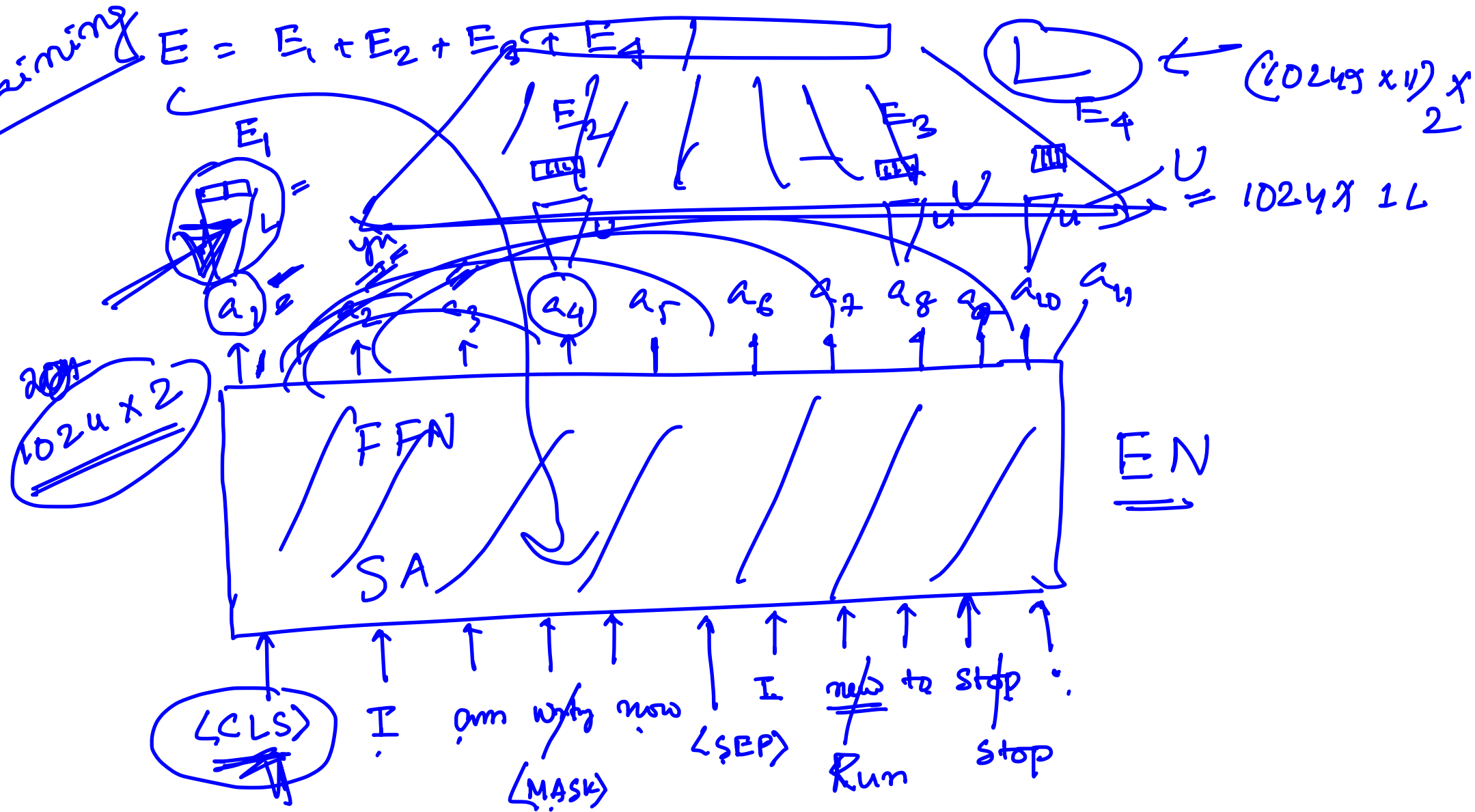


s_1 s_2
 s_i s_j



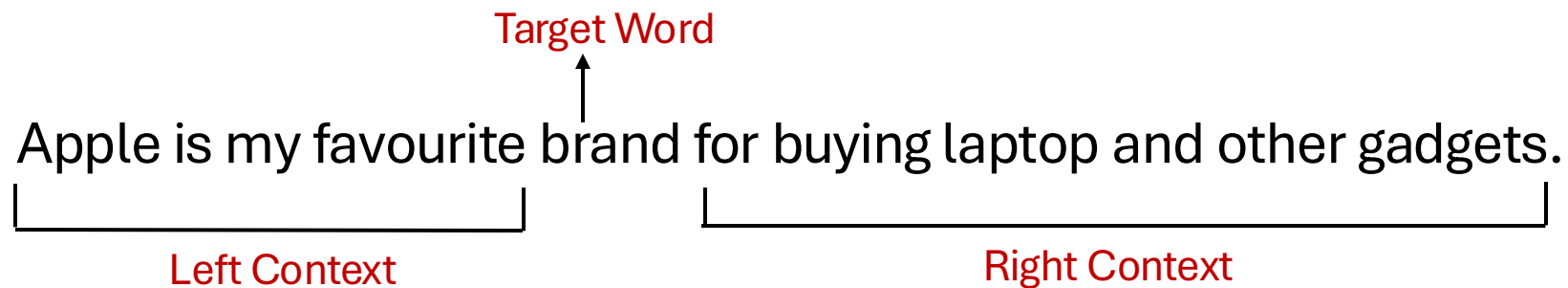
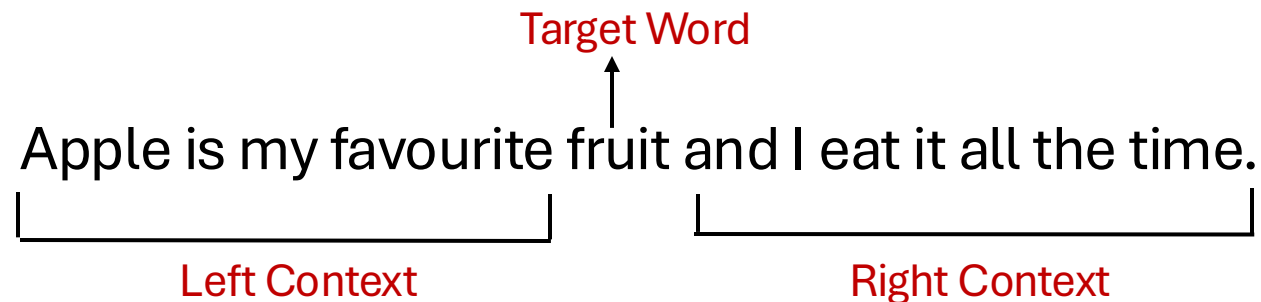


Pre-Training



Background - Bidirectional Context

- Bidirectional context, unlike unidirectional context, takes into account both the left and right contexts.



Motivation

- **Problem with previous methods:**
 - Language models only use left context or right context.
 - But language understanding is **bidirectional**.
- **Possible Issue:**
 - Directionality is needed to generate a well-formed probability distribution.
 - Words can see themselves in a bidirectional model.

Masked Language Modelling

- Mask out $k\%$ of the input words, and then predict the masked words (Usually $k = 15\%$). Example :

I like going to the [MASK] in the evening

↓
park

- Too little masking: Too expensive to train
- Too much masking: Not enough context
- The model needs to predict 15% of the words, but we don't replace with [MASK] 100% of the time. Instead:
 - 80% of the time, **replace with [MASK]** =
 - Example : like going to the park → like going to the [MASK]
 - 10% of the time, **replace random word**
 - Example : like going to the park → like going to the store
 - 10% of the time, **keep same**
 - Example : like going to the park → like going to the park

Next Sentence Prediction

- To learn relationships between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence.

Input = [CLS] I enjoy read [MASK] book ##s [SEP]

I finish ##ed a [MASK] novel [SEP]

Label = IsNext

Input = [CLS] I enjoy read ##ing book [MASK] [SEP]

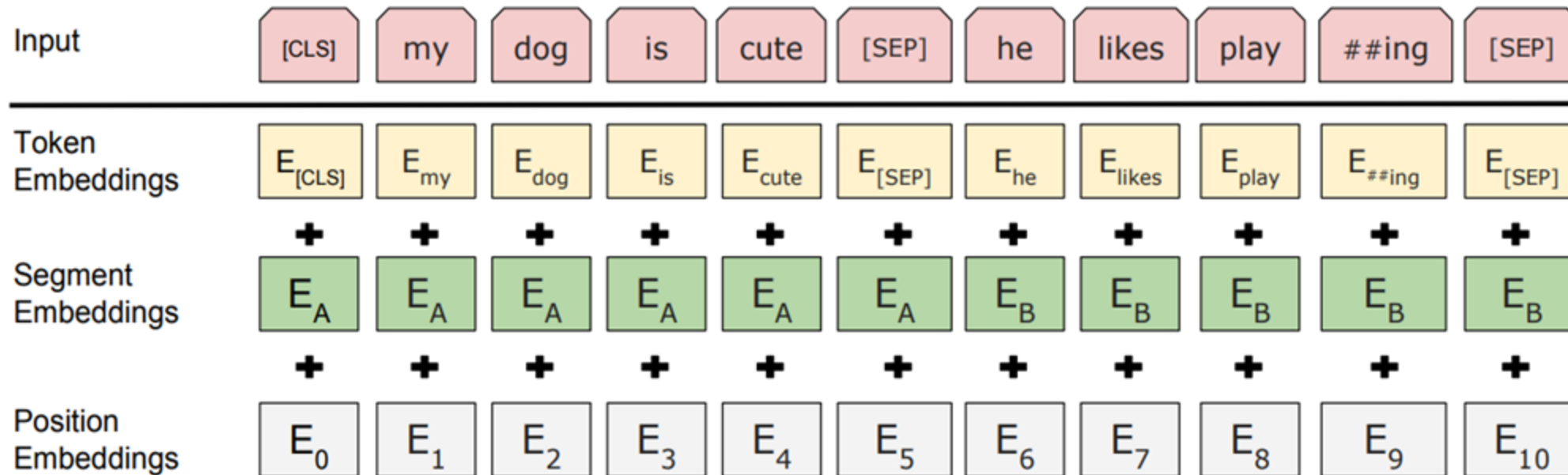
The dog ran [MASK] the street [SEP]

Label = NotNext

- Important for many important downstream tasks such as Question Answering (QA) and Natural Language Inference (NLI)
- How to choose sentences A and B for pretraining?
 - 50% of the time B is the actual next sentence that follows A (labeled as IsNext)
 - 50% of the time it is a random sentence from the corpus (labeled as NotNext)

Input Representation

- Use 30,000 WordPiece vocabulary on input.
- For a given token, its input representation is constructed by summing the token embeddings, the segmentation embeddings and the position embeddings.



Source of Image : BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (Devlin et al., NAACL 2019)